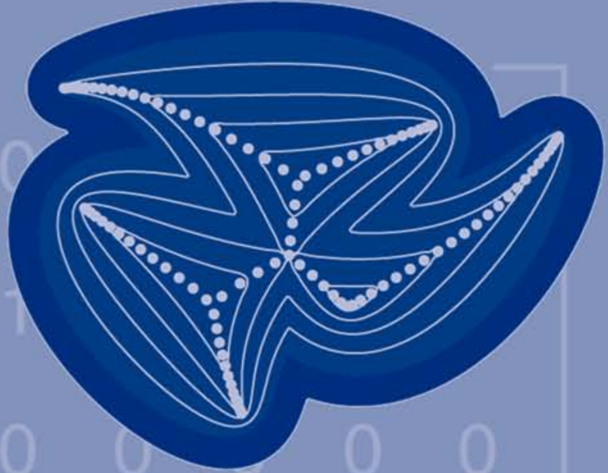
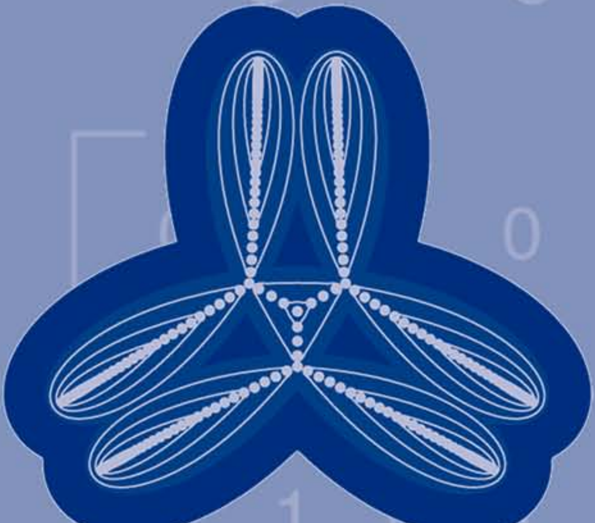
DISCRETE MATHEMATICS AND ITS APPLICATIONS
Series Editor KENNETH H. ROSEN

HANDBOOK OF LINEAR ALGEBRA





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EDITED BY
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HANDBOOK OF LINEAR ALGEBRA

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Dedication

I dedicate this book to my husband, Mark Hunacek, with gratitude both for his support throughout this project and for our wonderful life together.

Acknowledgments

I would like to thank Executive Editor Bob Stern of Taylor & Francis Group, who envisioned this project and whose enthusiasm and support has helped carry it to completion. I also want to thank Yolanda Croasdale, Suzanne Lassandro, Jim McGovern, Jessica Vakili and Mimi Williams, for their expert guidance of this book through the production process.

I would like to thank the many authors whose work appears in this volume for the contributions of their time and expertise to this project, and for their patience with the revisions necessary to produce a unified whole from many parts.

Without the help of the associate editors, Richard Brualdi, Anne Greenbaum, and Roy Mathias, this book would not have been possible. They gave freely of their time, expertise, friendship, and moral support, and I cannot thank them enough.

I thank Iowa State University for providing a collegial and supportive environment in which to work, not only during the preparation of this book, but for more than 25 years.

Leslie Hogben

The Editor

Leslie Hogben, Ph.D., is a professor of mathematics at Iowa State University. She received her B.A. from Swarthmore College in 1974 and her Ph.D. in 1978 from Yale University under the direction of Nathan Jacobson. Although originally working in nonassociative algebra, she changed her focus to linear algebra in the mid-1990s.

Dr. Hogben is a frequent organizer of meetings, workshops, and special sessions in combinatorial linear algebra, including the workshop, "Spectra of Families of Matrices Described by Graphs, Digraphs, and Sign Patterns," hosted by American Institute of Mathematics in 2006 and the Topics in Linear Algebra Conference hosted by Iowa State University in 2002. She is the Assistant Secretary/Treasurer of the International Linear Algebra Society.

An active researcher herself, Dr. Hogben particularly enjoys introducing graduate and undergraduate students to mathematical research. She has three current or former doctoral students and nine master's students, and has worked with many additional graduate students in the Iowa State University Combinatorial Matrix Theory Research Group, which she founded. Dr. Hogben is the co-director of the NSF-sponsored REU "Mathematics and Computing Research Experiences for Undergraduates at Iowa State University" and has served as a research mentor to ten undergraduates.

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Preface

It is no exaggeration to say that linear algebra is a subject of central importance in both mathematics and a variety of other disciplines. It is used by virtually all mathematicians and by statisticians, physicists, biologists, computer scientists, engineers, and social scientists. Just as the basic idea of first semester differential calculus (approximating the graph of a function by its tangent line) provides information about the function, the process of linearization often allows difficult problems to be approximated by more manageable linear ones. This can provide insight into, and, thanks to ever-more-powerful computers, approximate solutions of the original problem. For this reason, people working in all the disciplines referred to above should find the *Handbook of Linear Algebra* an invaluable resource.

The *Handbook* is the first resource that presents complete coverage of linear algebra, combinatorial linear algebra, and numerical linear algebra, combined with extensive applications to a variety of fields and information on software packages for linear algebra in an easy to use handbook format.

Content

The *Handbook* covers the major topics of linear algebra at both the graduate and undergraduate level as well as its offshoots (numerical linear algebra and combinatorial linear algebra), its applications, and software packages for linear algebra computations. The *Handbook* takes the reader from the very elementary aspects of the subject to the frontiers of current research, and its format (consisting of a number of independent chapters each organized in the same standard way) should make this book accessible to readers with divergent backgrounds.

Format

There are five main parts in this book. The first part (Chapters 1 through Chapter 26) covers linear algebra; the second (Chapter 27 through Chapter 36) and third (Chapter 37 through Chapter 49) cover, respectively, combinatorial and numerical linear algebra, two important branches of the subject. Applications of linear algebra to other disciplines, both inside and outside of mathematics, comprise the fourth part of the book (Chapter 50 through Chapter 70). Part five (Chapter 71 through Chapter 77) addresses software packages useful for linear algebra computations.

Each chapter is written by a different author or team of authors, who are experts in the area covered. Each chapter is divided into sections, which are organized into the following uniform format:

- Definitions
- Facts
- Examples

Most relevant definitions appear within the Definitions segment of each chapter, but some terms that are used throughout linear algebra are not redefined in each chapter. The Glossary, covering the terminology of linear algebra, combinatorial linear algebra, and numerical linear algebra, is available at the end of the book to provide definitions of terms that appear in different chapters. In addition to the definition, the Glossary also provides the number of the chapter (and section, thereof) where the term is defined. The Notation Index serves the same purpose for symbols.

The Facts (which elsewhere might be called theorems, lemmas, etc.) are presented in list format, which allows the reader to locate desired information quickly. In lieu of proofs, references are provided for all facts. The references will also, of course, supply a source of additional information about the subject of the chapter. In this spirit, we have encouraged the authors to use texts or survey articles on the subject as references, where available.

The Examples illustrate the definitions and facts. Each section is short enough that it is easy to go back and forth between the Definitions/Facts and the Examples to see the illustration of a fact or definition. Some sections also contain brief applications following the Examples (major applications are treated in their own chapters).

Feedback

To see updates and provide feedback and errata reports, please consult the web page for this book: http://www.public.iastate.edu/~lhogben/HLA.html or contact the editor via email, LHogben@iastate.edu, with *HLA* in the subject heading.

Preliminaries

This chapter contains a variety of definitions of terms that are used throughout the rest of the book, but are not part of linear algebra and/or do not fit naturally into another chapter. Since these definitions have little connection with each other, a different organization is followed; the definitions are (loosely) alphabetized and each definition is followed by an example.

Algebra

An **(associative) algebra** is a vector space A over a field F together with a **multiplication** $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{xy}$ from $A \times A$ to A satisfying two *distributive* properties and *associativity*, i.e., for all $a, b \in F$ and all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in A$:

$$(a\mathbf{x} + b\mathbf{y})\mathbf{z} = a(\mathbf{xz}) + b(\mathbf{yz}), \qquad \mathbf{x}(a\mathbf{y} + b\mathbf{z}) = a(\mathbf{xy}) + b(\mathbf{xz}) \qquad (\mathbf{xy})\mathbf{z} = \mathbf{x}(\mathbf{yz}).$$

Except in Chapter 69 and Chapter 70 the term *algebra* means associative algebra. In these two chapters, associativity is not assumed.

Examples:

The vector space of $n \times n$ matrices over a field F with matrix multiplication is an (associative) algebra.

Boundary

The **boundary** ∂S of a subset S of the real numbers or the complex numbers is the intersection of the closure of S and the closure of the complement of S.

Examples:

The boundary of $S = \{x \in \mathbb{C} : |z| \leq 1\}$ is $\partial S = \{x \in \mathbb{C} : |z| = 1\}$.

Complement

The **complement** of the set X in universe S, denoted $S \setminus X$, is all elements of S that are not in X. When the universe is clear (frequently the universe is $\{1, \ldots, n\}$) then this can be denoted X^c.

Examples:

For $S = \{1, 2, 3, 4, 5\}$ and $X = \{1, 3\}$, $S \setminus X = \{2, 4, 5\}$.

Complex Numbers

Let $a, b \in \mathbb{R}$. The symbol i denotes $\sqrt{-1}$.

The **complex conjugate** of a complex number $c = a + bi$ is $\overline{c} = a - bi$.
The **imaginary part** of $a + bi$ is $\mathrm{im}(a + bi) = b$ and the **real part** is $\mathrm{re}(a + bi) = a$.
The **absolute value** of $c = a + bi$ is $|c| = \sqrt{a^2 + b^2}$.

The **argument** of the nonzero complex number $re^{i\theta}$ is θ (with $r, \theta \in \mathbb{R}$ and $0 < r$ and $0 \leq \theta < 2\pi$).
The **open right half plane** $\mathbb{C}^+$ is $\{z \in \mathbb{C} : \mathrm{re}(z) > 0\}$.
The **closed right half plane** $\mathbb{C}_0^+$ is $\{z \in \mathbb{C} : \mathrm{re}(z) \geq 0\}$.
The **open left half plane** $\mathbb{C}^-$ is $\{z \in \mathbb{C} : \mathrm{re}(z) < 0\}$.
The **closed left half plane** $\mathbb{C}^-$ is $\{z \in \mathbb{C} : \mathrm{re}(z) \leq 0\}$.

Facts:

1. $|c| = c\overline{c}$
2. $|re^{i\theta}| = r$
3. $re^{i\theta} = r\cos\theta + r\sin\theta i$
4. $\overline{re^{i\theta}} = re^{-i\theta}$

Examples:

$\overline{2+3i} = 2 - 3i$, $\overline{1.4} = 1.4$, $1 + i = \sqrt{2}e^{i\pi/4}$.

Conjugate Partition

Let $\upsilon = (u_1, u_2, \ldots, u_n)$ be a sequence of integers such that $u_1 \geq u_2 \geq \cdots \geq u_n \geq 0$. The **conjugate partition** of υ is $\upsilon^* = (u_1^*, \ldots, u_t^*)$, where u_i^* is the number of js such that $u_j \geq i$. t is sometimes taken to be u_1, but is sometimes greater (obtained by extending with 0s).

Facts: If t is chosen to be the minimum, and $u_n > 0$, $\upsilon^{**} = \upsilon$.

Examples:

$(4, 3, 2, 2, 1)^* = (5, 4, 2, 1)$.

Convexity

Let V be a real or complex vector space.

Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\} \in V$. A vector of the form $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k$ with all the coefficients a_i nonnegative and $\sum a_i = 1$ is a **convex combination** of $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$.

A set $S \subseteq V$ is **convex** if any convex combination of vectors in S is in S.

The **convex hull** of S is the set of all convex combinations of S and is denoted by $\mathrm{Con}(S)$.

An **extreme point** of a closed convex set S is a point $\mathbf{v} \in S$ that is not a nontrivial convex combination of other points in S, i.e., $a\mathbf{x} + (1-a)\mathbf{y} = \mathbf{v}$ and $0 \leq a \leq 1$ implies $\mathbf{x} = \mathbf{y} = \mathbf{v}$.

A **convex polytope** is the convex hull of a finite set of vectors in $\mathbb{R}^n$.

Let $S \subseteq V$ be convex. A function $f : S \to \mathbb{R}$ is **convex** if for all $a \in \mathbb{R}, 0 < a < 1, \mathbf{x}, \mathbf{y} \in S$, $f(a\mathbf{x} + (1-a)\mathbf{y}) \leq af(\mathbf{x}) + (1-a)f(\mathbf{y})$.

Facts:

1. A set $S \subseteq V$ is convex if and only if $\mathrm{Con}(S) = S$.
2. The extreme points of $\mathrm{Con}(S)$ are contained in S.
3. [HJ85] *Krein-Milman Theorem*: A compact convex set is the convex hull of its extreme points.

Examples:

1. $[1.9, 0.8]^T$ is a convex combination of $[1, -1]^T$ and $[2, 1]^T$, since $[1.9, 0.8]^T = 0.1[1, -1]^T + 0.9[2, 1]^T$.
2. The set K of all $\mathbf{v} \in \mathbb{R}^3$ such that $v_i \geq 0, i = 1, 2, 3$ is a convex set. Its only extreme point is the zero vector.

Elementary Symmetric Function

The k**th elementary symmetric function of** $\alpha_i, i = 1, \ldots, n$ is

$$S_k(\alpha_1, \ldots, \alpha_n) = \sum_{1<i_1<i_2<\cdots<i_k<n} \alpha_{i_1}\alpha_{i_2}\ldots\alpha_{i_k}.$$

Examples:

$S_2(\alpha_1, \alpha_2, \alpha_3) = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3,$
$S_1(\alpha_1, \ldots, \alpha_n) = \alpha_1 + \alpha_2 + \cdots + \alpha_n$, $S_n(\alpha_1, \ldots, \alpha_n) = \alpha_1\alpha_2\ldots\alpha_n$.

Equivalence Relation

A binary relation $\equiv$ in a nonempty set S is an **equivalence relation** if it satisfies the following conditions:

1. (Reflexive) For all $a \in S, a \equiv a$.
2. (Symmetric) For all $a, b \in S, a \equiv b$ implies $b \equiv a$.
3. (Transitive) For all $a, b, c \in S, a \equiv b$ and $a \equiv b$ imply $a \equiv c$.

Examples:

Congruence mod n is an equivalence relation on the integers.

Field

A **field** is a set F with at least two elements together with a function $F \times F \to F$ called addition, denoted $(a, b) \to a + b$, and a function $F \times F \to F$ called multiplication, denoted $(a, b) \to ab$, which satisfy the following axioms:

1. (Commutativity) For each $a, b \in F, a + b = b + a$ and $ab = ba$.
2. (Associativity) For each $a, b, c \in F$, $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
3. (Identities) There exist two elements 0 and 1 in F such that $0 + a = a$ and $1a = a$ for each $a \in F$.
4. (Inverses) For each $a \in F$, there exists an element $-a \in F$ such that $(-a) + a = 0$. For each nonzero $a \in F$, there exists an element $a^{-1} \in F$ such that $a^{-1}a = 1$.
5. (Distributivity) For each $a, b, c \in F, a(b + c) = ab + ac$.

Examples:

The real numbers, $\mathbb{R}$, the complex numbers, $\mathbb{C}$, and the rational numbers, $\mathbb{Q}$, are all fields. The set of integers, $\mathbb{Z}$, is not a field.

Greatest Integer Function

The **greatest integer** or **floor** function $\lfloor x \rfloor$ (defined on the real numbers) is the greatest integer less than or equal to x.

Examples:

$\lfloor 1.5 \rfloor = 1$, $\lfloor 1 \rfloor = 1$, $\lfloor -1.5 \rfloor = -2$.

Group

(See also Chapter 67 and Chapter 68.)

A **group** is a nonempty set G with a function $G \times G \to G$ denoted $(a, b) \to ab$, which satisfies the following axioms:

1. (Associativity) For each $a, b, c \in G$, $(ab)c = a(bc)$.
2. (Identity) There exists an element $e \in G$ such that $ea = a = ae$ for each $a \in G$.
3. (Inverses) For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a^{-1}a = e = aa^{-1}$.

A group is **abelian** if $ab = ba$ for all $a, b \in G$.

Examples:

1. Any vector space is an abelian group under $+$.
2. The set of invertible $n \times n$ real matrices is a group under matrix multiplication.
3. The set of all permutations of a set is a group under composition.

Interlaces

Let $a_1 \geq a_2 \geq \cdots \geq a_n$ and $b_1 \geq b_2 \geq \cdots \geq b_{n-1}$, two sequences of real numbers arranged in decreasing order. Then the sequence $\{b_i\}$ **interlaces** the sequence $\{a_i\}$ if $a_n \leq b_{n-1} \leq a_{n-1} \cdots \leq b_1 \leq a_1$. Further, if all of the above inequalities can be taken to be strict, the sequence $\{b_i\}$ **strictly interlaces** the sequence $\{a_i\}$. Analogous definitions are given when the numbers are in increasing order.

Examples:

$7 \geq 2.2 \geq -1$ strictly interlaces $11 \geq \pi \geq 0 \geq -2.6$.

Majorization

Let $\alpha = (a_1, a_2, \ldots, a_n)$, $\beta = (b_1, b_2, \ldots, b_n)$ be sequences of real numbers.

$\alpha^{\downarrow} = (a_1^{\downarrow}, a_2^{\downarrow}, \ldots, a_n^{\downarrow})$ is the permutation of α with entries in nonincreasing order, i.e., $a_1^{\downarrow} \geq a_2^{\downarrow} \geq \ldots \geq a_n^{\downarrow}$.

$\alpha^{\uparrow} = (a_1^{\uparrow}, a_2^{\uparrow}, \ldots, a_n^{\uparrow})$ is the permutation of α with entries in nondecreasing order, i.e., $a_1^{\uparrow} \leq a_2^{\uparrow} \leq \ldots \leq a_n^{\uparrow}$.

α **weakly majorizes** β, written $\alpha \succeq_w \beta$ or $\beta \preceq_w \alpha$, if:

$$\sum_{i=1}^{k} a_i^{\downarrow} \geq \sum_{i=1}^{k} b_i^{\downarrow} \quad \text{for all } k = 1, \ldots n.$$

α **majorizes** β, written $\alpha \succeq \beta$ or $\beta \preceq \alpha$, if $\alpha \succeq_w \beta$ and $\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} b_i$.

Examples:

1. If $\alpha = (2, 2, -1.3, 8, 7.7)$, then $\alpha^{\downarrow} = (8, 7.7, 2, 2, -1.3)$ and $\alpha^{\uparrow} = (-1.3, 2, 2, 7.7, 8)$.
2. $(5,3,1.5,1.5,1) \succeq (4,3,2,2,1)$ and $(6,5,0) \succeq_w (4,3,2)$.

Metric

A **metric** on a set S is a real-valued function $f : S \times S \to \mathbb{R}$ satisfying the following conditions:

1. For all $x, y \in S$, $f(x, y) \geq 0$.
2. For all $x \in S$, $f(x, x) = 0$.
3. For all $x, y \in S$, $f(x, y) = 0$ implies $x = y$.
4. For all $x, y \in S$, $f(x, y) = f(y, x)$.
5. For all $x, y, z \in S$, $f(x, y) + f(y, z) \geq f(x, z)$.

A metric is intended as a measure of distance between elements of the set.

Examples:

If $\|\cdot\|$ is a norm on a vector space, then $f(x, y) = \|\mathbf{x} - \mathbf{y}\|$ is a metric.

Multiset

A **multiset** is an unordered list of elements that allows repetition.

Examples:

Any set is a multiset, but $\{1, 1, 3, -2, -2, -2\}$ is a multiset that is not a set.

O and o

Let, f, g be real valued functions of $\mathbb{N}$ or $\mathbb{R}$, i.e., $f, g : \mathbb{N} \to \mathbb{R}$ or $f, g : \mathbb{R} \to \mathbb{R}$.

f is $O(g)$ (**big-oh** of g) if there exist constants C, k such that $|f(x)| \leq C|g(x)|$ for all $x \geq k$.

f is $o(g)$ (**little-oh** of g) if $\lim_{x\to\infty} \left|\frac{f(n)}{g(n)}\right| = 0$.

Examples:

$x^2 + x$ is $O(x^2)$ and $\ln x$ is $o(x)$.

Path-connected

A subset S of the complex numbers is **path-connected** if for any $x, y \in S$ there exists a continuous function $p : [0, 1] \to S$ with $p(0) = x$ and $p(1) = y$.

Examples:

$S = \{z \in \mathbb{C} : 1 \leq |z| \leq 2\}$ and the line $\{a + bi : a = 2b + 3\}$ are path-connected.

Permutations

A **permutation** is a one-to-one onto function from a set to itself.

The set of permutations of $\{1, \ldots, n\}$ is denoted S_n. The identity permutation is denoted ε_n. In this book, permutations are generally assumed to be elements of S_n for some n.

A **cycle** or k-**cycle** is a permutation τ such that there is a subset $\{a_1, \ldots, a_k\}$ of $\{1, \ldots, n\}$ satisfying $\tau(a_i) = a_{i+1}$ and $\tau(a_k) = a_1$; this is denoted $\tau = (a_1, a_2, \ldots, a_k)$. The **length** of this cycle is k.

A **transposition** is a 2-cycle.

A permutation is **even** (respectively, **odd**) if it can be written as the product of an even (odd) number of transpositions.

The **sign** of a permutation τ, denoted sgn τ, is +1 if τ is even and -1 if τ is odd.

Note: Permutations are functions and act from the left (see Examples).

Facts:

1. Every permutation can be expressed as a product of disjoint cycles. This expression is unique up to the order of the cycles in the decomposition and cyclic permutation within a cycle.
2. Every permutation can be written as a product of transpositions. If some such expression includes an even number of transpositions, then every such expression includes an even number of transpositions.
3. S_n with the operation of composition is a group.

Examples:

1. If $\tau = (1523) \in S_6$, then $\tau(1) = 5, \tau(2) = 3, \tau(3) = 1, \tau(4) = 4, \tau(5) = 2, \tau(6) = 6$.
2. (123)(12)=(13).
3. sgn(1234) = −1, because (1234) = (14)(13)(12).

Ring

(See also Section 23.1)

A **ring** is a set R together with a function $R \times R \to R$ called addition, denoted $(a, b) \to a + b$, and a function $R \times R \to R$ called multiplication, denoted $(a, b) \to ab$, which satisfy the following axioms:

1. (Commutativity of +) For each $a, b \in R$, $a + b = b + a$.
2. (Associativity) For each $a, b, c \in R$, $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
3. (+ identity) There exists an element 0 in R such that $0 + a = a$.
4. (+ inverse) For each $a \in R$, there exists an element $-a \in R$ such that $(-a) + a = 0$.
5. (Distributivity) For each $a, b, c \in R, a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

A **zero divisor** in a ring R is a nonzero element $a \in R$ such that there exists a nonzero $b \in R$ with $ab = 0$ or $ba = 0$.

Examples:

- The set of integers, $\mathbb{Z}$, is a ring.
- Any field is a ring.
- Let F be a field. Then $F^{n\times n}$, with matrix addition and matrix multiplication as the operations, is a ring. $E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ are zero divisors since $E_{11}E_{22} = 0_2$.

Sign

(For sign of a permutation, see *permutation*.)

The **sign** of a complex number is defined by:

$$\operatorname{sign}(z) = \begin{cases} z/|z|, & \text{if } z \neq 0; \\ 1, & \text{if } z = 0. \end{cases}$$

If z is a real number, this sign function yields 1 or −1.

This sign function is used in numerical linear algebra.

The **sign** of a real number (as used in sign patterns) is defined by:

$$\operatorname{sgn}(a) = \begin{cases} +, & \text{if } a > 0; \\ 0, & \text{if } a = 0; \\ -, & \text{if } a < 0. \end{cases}$$

This sign function is used in combinatorial linear algebra, and the product of a sign and a real number is interpreted in the obvious way as a real number.

Warning: The two sign functions disagree on the sign of 0.

Examples:

$\operatorname{sgn}(-1.3) = -$, $\operatorname{sign}(-1.3) = -1$, $\operatorname{sgn}(0) = 0$, $\operatorname{sign}(0) = 1$,
$\operatorname{sign}(1 + i) = \frac{(1 + i)}{\sqrt{2}}$.

References

[HJ85] [HJ85] R. Horn and C. R. Johnson. *Matrix Analysis.* Cambridge University Press, Cambridge, 1985.

I

Linear Algebra

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Basic Linear Algebra

1

Vectors, Matrices, and Systems of Linear Equations

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Throughout this chapter, F will denote a field. The references [Lay03], [Leo02], and [SIF00] are good sources for more detail about much of the material in this chapter. They discuss primarily the field of real numbers, but the proofs are usually valid for any field.

1.1 Vector Spaces

Vectors are used in many applications. They often represent quantities that have both direction and magnitude, such as velocity or position, and can appear as functions, as n-tuples of scalars, or in other disguises. Whenever objects can be added and multiplied by scalars, they may be elements of some vector space. In this section, we formulate a general definition of vector space and establish its basic properties. An element of a field, such as the real numbers or the complex numbers, is called a scalar to distinguish it from a vector.

Definitions:

A **vector space over** F is a set V together with a function $V \times V \to V$ called **addition**, denoted $(\mathbf{x},\mathbf{y}) \to \mathbf{x} + \mathbf{y}$, and a function $F \times V \to V$ called **scalar multiplication** and denoted $(c,\mathbf{x}) \to c\mathbf{x}$, which satisfy the following axioms:

1. (Commutativity) For each $\mathbf{x}, \mathbf{y} \in V$, $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$.
2. (Associativity) For each $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$, $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$.
3. (Additive identity) There exists a **zero vector** in V, denoted $\mathbf{0}$, such that $\mathbf{0} + \mathbf{x} = \mathbf{x}$ for each $\mathbf{x} \in V$.
4. (Additive inverse) For each $\mathbf{x} \in V$, there exists $-\mathbf{x} \in V$ such that $(-\mathbf{x}) + \mathbf{x} = \mathbf{0}$.
5. (Distributivity) For each $a \in F$ and $\mathbf{x}, \mathbf{y} \in V$, $a(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + a\mathbf{y}$.
6. (Distributivity) For each $a, b \in F$ and $\mathbf{x} \in V$, $(a + b)\,\mathbf{x} = a\mathbf{x} + b\mathbf{x}$.

7. (Associativity) For each $a, b \in F$ and $\mathbf{x} \in V$, $(ab)\,\mathbf{x} = a(b\mathbf{x})$.
8. For each $\mathbf{x} \in V$, $1\mathbf{x} = \mathbf{x}$.

The properties that for all $\mathbf{x}, \mathbf{y} \in V$, and $a \in F$, $\mathbf{x}+\mathbf{y} \in V$ and $a\mathbf{x} \in V$, are called **closure under addition** and **closure under scalar multiplication**, respectively. The elements of a vector space V are called **vectors**. A vector space is called **real** if $F = \mathbb{R}$, **complex** if $F = \mathbb{C}$.

If n is a positive integer, F^n denotes the set of all ordered n-tuples (written as columns). These are sometimes written instead as rows $[x_1 \quad \cdots \quad x_n]$ or $(x_1, \ldots, x_n)$. For $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in F^n$ and $c \in F$, define addition and scalar multiplication coordinate-wise: $\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$ and $c\mathbf{x} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}$. Let $\mathbf{0}$ denote the n-tuple of zeros. For $\mathbf{x} \in F^n$, x_j is called the j^{th} **coordinate** of $\mathbf{x}$.

A **subspace** of vector space V over field F is a subset of V, which is itself a vector space over F when the addition and scalar multiplication of V are used. If S_1 and S_2 are subsets of vector space V, define $S_1 + S_2 = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in S_1 \text{ and } \mathbf{y} \in S_2\}$.

Facts:

Let V be a vector space over F.

1. F^n is a vector space over F.
2. [FIS03, pp. 11–12] (Basic properties of a vector space):
   - The vector $\mathbf{0}$ is the only additive identity in V.
   - For each $\mathbf{x} \in V$, $-\mathbf{x}$ is the only additive inverse for $\mathbf{x}$ in V.
   - For each $\mathbf{x} \in V$, $-\mathbf{x} = (-1)\mathbf{x}$.
   - If $a \in F$ and $\mathbf{x} \in V$, then $a\mathbf{x} = \mathbf{0}$ if and only if $a = 0$ or $\mathbf{x} = \mathbf{0}$.
   - (Cancellation) If $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $\mathbf{x} + \mathbf{y} = \mathbf{x} + \mathbf{z}$, then $\mathbf{y} = \mathbf{z}$.
3. [FIS03, pp. 16–17] Let W be a subset of V. The following are equivalent:
   - W is a subspace of V.
   - W is nonempty and closed under addition and scalar multiplication.
   - $\mathbf{0} \in W$ and for any $\mathbf{x}, \mathbf{y} \in W$ and $a, b \in F$, $a\mathbf{x} + b\mathbf{y} \in W$.
4. For any vector space V, $\{\mathbf{0}\}$ and V itself are subspaces of V.
5. [FIS03, p. 19] The intersection of any nonempty collection of subspaces of V is a subspace of V.
6. [FIS03, p. 22] Let W_1 and W_2 be subspaces of V. Then $W_1 + W_2$ is a subspace of V containing W_1 and W_2. It is the smallest subspace that contains them in the sense that any subspace that contains both W_1 and W_2 must contain $W_1 + W_2$.

Examples:

1. The set $\mathbb{R}^n$ of all ordered n-tuples of real numbers is a vector space over $\mathbb{R}$, and the set $\mathbb{C}^n$ of all ordered n-tuples of complex numbers is a vector space over $\mathbb{C}$. For instance, $\mathbf{x} = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 2i \\ 4 \\ 2 - 3i \end{bmatrix}$ are elements of $\mathbb{C}^3$; $\mathbf{x} + \mathbf{y} = \begin{bmatrix} 3 + 2i \\ 4 \\ 1 - 3i \end{bmatrix}$, $-\mathbf{y} = \begin{bmatrix} -2i \\ -4 \\ -2 + 3i \end{bmatrix}$, and $i\mathbf{y} = \begin{bmatrix} -2 \\ 4i \\ 3 + 2i \end{bmatrix}$.
2. Notice $\mathbb{R}^n$ is a subset of $\mathbb{C}^n$ but not a subspace of $\mathbb{C}^n$, since $\mathbb{R}^n$ is not closed under multiplication by nonreal numbers.

3. The vector spaces $\mathbb{R}$, $\mathbb{R}^2$, and $\mathbb{R}^3$ are the usual Euclidean spaces of analytic geometry. There are three types of subspaces of $\mathbb{R}^2$: $\{\mathbf{0}\}$, a line through the origin, and $\mathbb{R}^2$ itself. There are four types of subspaces of $\mathbb{R}^3$: $\{\mathbf{0}\}$, a line through the origin, a plane through the origin, and $\mathbb{R}^3$ itself. For instance, let $\mathbf{v} = (5, -1, -1)$ and $\mathbf{w} = (0, 3, -2)$. The lines $W_1 = \{s\mathbf{v} : s \in \mathbb{R}\}$ and $W_2 = \{s\mathbf{w} : s \in \mathbb{R}\}$ are subspaces of $\mathbb{R}^3$. The subspace $W_1 + W_2 = \{s\mathbf{v} + t\,\mathbf{w} : s, t \in \mathbb{R}\}$ is a plane. The set $\{s\mathbf{v} + \mathbf{w} : s \in \mathbb{R}\}$ is a line parallel to W_1, but is not a subspace. (For more information on geometry, see Chapter 65.)
4. Let $F[x]$ be the set of all polynomials in the single variable x, with coefficients from F. To add polynomials, add coefficients of like powers; to multiply a polynomial by an element of F, multiply each coefficient by that scalar. With these operations, $F[x]$ is a vector space over F. The zero polynomial z, with all coefficients 0, is the additive identity of $F[x]$. For $f \in F[x]$, the function $-f$ defined by $-f(x) = (-1)f(x)$ is the additive inverse of f.
5. In $F[x]$, the constant polynomials have degree 0. For $n > 0$, the polynomials with highest power term x^n are said to have degree n. For a nonnegative integer n, let $F[x; n]$ be the subset of $F[x]$ consisting of all polynomials of degree n or less. Then $F[x; n]$ is a subspace of $F[x]$.
6. When $n > 0$, the set of all polynomials of degree exactly n is not a subspace of $F[x]$ because it is not closed under addition or scalar multiplication. The set of all polynomials in $\mathbb{R}[x]$ with rational coefficients is not a subspace of $\mathbb{R}[x]$ because it is not closed under scalar multiplication.
7. Let V be the set of all infinite sequences $(a_1, a_2, a_3, \ldots)$, where each $a_j \in F$. Define addition and scalar multiplication coordinate-wise. Then V is a vector space over F.
8. Let X be a nonempty set and let $\mathcal{F}(X, F)$ be the set of all functions $f : X \to F$. Let $f, g \in \mathcal{F}(X, F)$ and define $f + g$ and cf pointwise, as $(f + g)(x) = f(x) + g(x)$ and $(cf)(x) = cf(x)$ for all $x \in X$. With these operations, $\mathcal{F}(X, F)$ is a vector space over F. The zero function is the additive identity and $(-1)f = -f$, the additive inverse of f.
9. Let X be a nonempty subset of $\mathbb{R}^n$. The set $C(X)$ of all continuous functions $f : X \to \mathbb{R}$ is a subspace of $\mathcal{F}(X, \mathbb{R})$. The set $\mathcal{D}(X)$ of all differentiable functions $f : X \to \mathbb{R}$ is a subspace of $C(X)$ and also of $\mathcal{F}(X, \mathbb{R})$.

1.2 Matrices

Matrices are rectangular arrays of scalars that are used in a great variety of ways, such as to solve linear systems, model linear behavior, and approximate nonlinear behavior. They are standard tools in almost every discipline, from sociology to physics and engineering.

Definitions:

An $m \times p$ **matrix** over F is an $m \times p$ rectangular array $A = \begin{bmatrix} a_{11} & \cdots & a_{1p} \\ \vdots & \cdots & \vdots \\ a_{m1} & \cdots & a_{mp} \end{bmatrix}$, with entries from F. The notation $A = [a_{ij}]$ that displays a typical entry is also used. The element a_{ij} of the matrix A is called the (i, j) **entry** of A and can also be denoted $(A)_{ij}$. The **shape** (or **size**) of A is $m \times p$, and A is **square** if $m = p$; in this case, m is also called the size of A. Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be **equal** if they have the same shape and $a_{ij} = b_{ij}$ for all i, j. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times p$ matrices, and let c be a scalar. Define **addition** and **scalar multiplication** on the set of all $m \times p$ matrices over F entrywise, as $A + B = [a_{ij} + b_{ij}]$ and $cA = [ca_{ij}]$. The set of all $m \times p$ matrices over F with these operations is denoted $F^{m \times p}$.

If A is $m \times p$, **row** i is $[a_{i1}, \ \ldots, \ a_{ip}]$ and **column** j is $\begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix}$. These are called a **row vector** and a **column vector** respectively, and they belong to $F^{n \times 1}$ and $F^{1 \times n}$, respectively. The elements of F^n are identified with the elements of $F^{n \times 1}$ (or sometimes with the elements of $F^{1 \times n}$). Let $\mathbf{0}_{mp}$ denote the $m \times p$ matrix of zeros, often shortened to $\mathbf{0}$ when the size is clear. Define $-A = (-1)A$.

Let $A = [\mathbf{a}_1 \quad \dots \quad \mathbf{a}_p] \in F^{m\times p}$, where $\mathbf{a}_j$ is the jth column of A, and let $\mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_p \end{bmatrix} \in F^{p\times 1}$. The **matrix–vector product** of A and $\mathbf{b}$ is $A\mathbf{b} = b_1\mathbf{a}_1 + \dots + b_p\mathbf{a}_p$. Notice $A\mathbf{b}$ is $m \times 1$.

If $A \in F^{m\times p}$ and $C = [\mathbf{c}_1 \quad \dots \quad \mathbf{c}_n] \in F^{p\times n}$, define the **matrix product** of A and C as $AC = [A\mathbf{c}_1 \quad \dots \quad A\mathbf{c}_n]$. Notice AC is $m \times n$.

Square matrices A and B **commute** if $AB = BA$. When $i = j$, a_{ii} is a **diagonal entry** of A and the set of all its diagonal entries is the **main diagonal** of A. When $i \neq j$, a_{ij} is an **off-diagonal entry**.

The **trace** of A is the sum of all the diagonal entries of A, $\operatorname{tr} A = \sum_{i=1}^{n} a_{ii}$.

A matrix $A = [a_{ij}]$ is **diagonal** if $a_{ij} = 0$ whenever $i \neq j$, **lower triangular** if $a_{ij} = 0$ whenever $i < j$, and **upper triangular** if $a_{ij} = 0$ whenever $i > j$. A **unit triangular** matrix is a lower or upper triangular matrix in which each diagonal entry is 1.

The **identity matrix** I_n, often shortened to I when the size is clear, is the $n \times n$ matrix with main diagonal entries 1 and other entries 0.

A **scalar matrix** is a scalar multiple of the identity matrix.

A **permutation matrix** is one whose rows are some rearrangement of the rows of an identity matrix.

Let $A \in F^{m\times p}$. The **transpose** of A, denoted A^T, is the $p \times m$ matrix whose (i, j) entry is the (j, i) entry of A.

The square matrix A is **symmetric** if $A^T = A$ and **skew-symmetric** if $A^T = -A$.

When $F = \mathbb{C}$, that is, when A has complex entries, the **Hermitian adjoint** of A is its conjugate transpose, $A^* = \bar{A}^T$; that is, the (i, j) entry of A^* is $\overline{a_{ji}}$. Some authors, such as [Leo02], write A^H instead of A^*.

The square matrix A is **Hermitian** if $A^* = A$ and **skew-Hermitian** if $A^* = -A$.

Let α be a nonempty set of row indices and β a nonempty set of column indices. A **submatrix** of A is a matrix $A[\alpha, \beta]$ obtained by choosing the entries of A, which lie in rows α and columns β. A **principal submatrix** of A is a submatrix of the form $A[\alpha, \alpha]$. A **leading principal submatrix** of A is one of the form $A[\{1, \dots, k\}, \{1, \dots, k\}]$.

Facts:

1. [SIF00, p. 5] $F^{m\times p}$ is a vector space over F. That is, if $\mathbf{0}$, A, B, $C \in F^{m\times p}$, and c,$d \in F$, then:
   - $A + B = B + A$
   - $(A + B) + C = A + (B + C)$
   - $A + \mathbf{0} = \mathbf{0} + A = A$
   - $A + (-A) = (-A) + A = \mathbf{0}$
   - $c(A + B) = cA + cB$
   - $(c + d)A = cA + dA$
   - $(cd)\, A = c(dA)$
   - $1A = A$
2. If $A \in F^{m\times p}$ and $C \in F^{p\times n}$, the (i, j) entry of AC is $(AC)_{ij} = \sum_{k=1}^{p} a_{ik}a_{kj}$. This is the matrix product of row i of A and column j of C.
3. [SIF00, p. 88] Let $c \in F$, let A and B be matrices over F, let I denote an identity matrix, and assume the shapes allow the following sums and products to be calculated. Then:
   - $AI = IA = A$
   - $A\mathbf{0} = \mathbf{0}$ and $\mathbf{0}A = \mathbf{0}$
   - $A(BC) = (AB)C$
   - $A(B + C) = AB + AC$
   - $(A + B)C = AC + BC$
   - $c(AB) = A(cB) = (cA)B$ for any scalar c

4. [SIF00, p. 5 and p. 20] Let $c \in F$, let A and B be matrices over F, and assume the shapes allow the following sums and products to be calculated. Then:
   - $(A^T)^T = A$
   - $(A+B)^T = A^T + B^T$
   - $(cA)^T = cA^T$
   - $(AB)^T = B^T A^T$
5. [Leo02, pp. 321–323] Let $c \in \mathbb{C}$, let A and B be matrices over $\mathbb{C}$, and assume the shapes allow the following sums and products to be calculated. Then:
   - $(A^*)^* = A$
   - $(A+B)^* = A^* + B^*$
   - $(cA)^* = \bar{c}A^*$
   - $(AB)^* = B^* A^*$
6. If A and B are $n \times n$ and upper (lower) triangular, then AB is upper (lower) triangular.

Examples:

1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 7 \\ 8 \\ -9 \end{bmatrix}$. By definition, $A\mathbf{b} = 7\begin{bmatrix} 1 \\ 4 \end{bmatrix} + 8\begin{bmatrix} 2 \\ 5 \end{bmatrix} - 9\begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} -4 \\ 14 \end{bmatrix}$. Hand calculation of $A\mathbf{b}$ can be done more quickly using Fact 2: $A\mathbf{b} = \begin{bmatrix} 1\cdot 7 + 2\cdot 8 - 3\cdot 9 \\ 4\cdot 7 + 5\cdot 8 - 6\cdot 9 \end{bmatrix} = \begin{bmatrix} -4 \\ 14 \end{bmatrix}$.
2. Let $A = \begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 8 \end{bmatrix}$, $B = \begin{bmatrix} -3 & 8 & 0 \\ 1 & 2 & -5 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & -1 & 8 \\ 1 & 3 & 0 \\ 1 & 2 & -2 \end{bmatrix}$. Then $A + B = \begin{bmatrix} -2 & 5 & 4 \\ 3 & 2 & 3 \end{bmatrix}$ and $2A = \begin{bmatrix} 2 & -6 & 8 \\ 4 & 0 & 16 \end{bmatrix}$. The matrices $A + C$, BA, and AB are not defined, but
$$AC = \left[A\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad A\begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} \quad A\begin{bmatrix} 8 \\ 0 \\ -2 \end{bmatrix} \right] = \begin{bmatrix} 2 & -2 & 0 \\ 10 & 14 & 0 \end{bmatrix}.$$
3. Even when the shapes of A and B allow both AB and BA to be calculated, AB and BA are not usually equal. For instance, let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$; then $AB = \begin{bmatrix} a & b \\ 2c & 2d \end{bmatrix}$ and $BA = \begin{bmatrix} a & 2b \\ c & 2d \end{bmatrix}$, which will be equal only if $b = c = 0$.
4. The product of matrices can be a zero matrix even if neither has any zero entries. For example, if $A = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, then $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. Notice that BA is also defined but has no zero entries: $BA = \begin{bmatrix} 3 & -3 \\ 3 & -3 \end{bmatrix}$.
5. The matrices $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -9 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \end{bmatrix}$ are diagonal, $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 5 & -9 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 0 \end{bmatrix}$ are lower triangular, and $\begin{bmatrix} 1 & -4 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & -9 \end{bmatrix}$ and $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ are upper triangular. The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 5 & 1 \end{bmatrix}$ is unit lower triangular, and its transpose is unit upper triangular.

6. Examples of permutation matrices include every identity matrix, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, and $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

7. Let $A = \begin{bmatrix} 1+i & -3i & 4 \\ 1+2i & 5i & 0 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 1+i & 1+2i \\ -3i & 5i \\ 4 & 0 \end{bmatrix}$ and $A^* = \begin{bmatrix} 1-i & 1-2i \\ 3i & -5i \\ 4 & 0 \end{bmatrix}$.

8. The matrices $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and $\begin{bmatrix} i & 2 & 3+2i \\ 2 & 4-i & 5i \\ 3+2i & 5i & 6 \end{bmatrix}$ are symmetric.

9. The matrices $\begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 2 & 3+2i \\ -2 & 0 & -5 \\ -3-2i & 5 & 0 \end{bmatrix}$ are skew-symmetric.

10. The matrix $\begin{bmatrix} 1 & 2+i & 1-3i \\ 2-i & 0 & 1 \\ 1+3i & 1 & 6 \end{bmatrix}$ is Hermitian, and any real symmetric matrix, such as $\begin{bmatrix} 4 & 2 & 3 \\ 2 & 0 & 5 \\ 3 & 5 & -1 \end{bmatrix}$, is also Hermitian.

11. The matrix $\begin{bmatrix} i & 2 & -3+2i \\ -2 & 4i & 5 \\ 3+2i & -5 & 0 \end{bmatrix}$ is skew-Hermitian, and any real skew-symmetric matrix, such as $\begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}$, is also skew-Hermitian.

12. Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$. Row 1 of A is $[1 \quad 2 \quad 3 \quad 4]$, column 3 is $\begin{bmatrix} 3 \\ 7 \\ 11 \\ 15 \end{bmatrix}$, and the submatrix in rows $\{1, 2, 4\}$ and columns $\{2, 3, 4\}$ is $A[\{1,2,4\},\{2,3,4\}] = \begin{bmatrix} 2 & 3 & 4 \\ 6 & 7 & 8 \\ 14 & 15 & 16 \end{bmatrix}$. A principal submatrix of A is $A[\{1, 2, 4\}, \{1, 2, 4\}] = \begin{bmatrix} 1 & 2 & 4 \\ 5 & 6 & 8 \\ 13 & 14 & 16 \end{bmatrix}$. The leading principal submatrices of A are $[1]$, $\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \\ 9 & 10 & 11 \end{bmatrix}$, and A itself.

1.3 Gaussian and Gauss–Jordan Elimination

Definitions:

Let A be a matrix with m rows.

When a row of A is not zero, its first nonzero entry is the **leading entry** of the row. The matrix A is in **row echelon form** (REF) when the following two conditions are met:

1. Any zero rows are below all nonzero rows.
2. For each nonzero row $i, i \leq m-1$, either row $i+1$ is zero or the leading entry of row $i+1$ is in a column to the right of the column of the leading entry in row i.

The matrix A is in **reduced row echelon form** (RREF) if it is in row echelon form and the following third condition is also met:

3. If a_{ik} is the leading entry in row i, then $a_{ik} = 1$, and every entry of column k other than a_{ik} is zero.

Elementary row operations on a matrix are operations of the following types:

1. Add a multiple of one row to a different row.
2. Exchange two different rows.
3. Multiply one row by a nonzero scalar.

The matrix A is **row equivalent** to the matrix B if there is a sequence of elementary row operations that transforms A into B. The **reduced row echelon form** of A, RREF(A), is the matrix in reduced row echelon form that is row equivalent to A. A **row echelon form of** A is any matrix in row echelon form that is row equivalent to A. The **rank** of A, denoted rank A or rank(A), is the number of leading entries in RREF(A). If A is in row echelon form, the positions of the leading entries in its nonzero rows are called **pivot positions** and the entries in those positions are called **pivots**. A column (row) that contains a pivot position is a **pivot column (pivot row)**.

Gaussian Elimination is a process that uses elementary row operations in a particular way to change, or reduce, a matrix to row echelon form. **Gauss–Jordan Elimination** is a process that uses elementary row operations in a particular way to reduce a matrix to RREF. See Algorithm 1 below.

Facts:

Let $A \in F^{m \times p}$.

1. [Lay03, p. 15] The reduced row echelon form of A, RREF(A), exists and is unique.
2. A matrix in REF or RREF is upper triangular.
3. Every elementary row operation is reversible by an elementary row operation of the same type.
4. If A is row equivalent to B, then B is row equivalent to A.
5. If A is row equivalent to B, then RREF(A) = RREF(B) and rank A = rank B.
6. The number of nonzero rows in any row echelon form of A equals rank A.
7. If B is any row echelon form of A, the positions of the leading entries in B are the same as the positions of the leading entries of RREF(A).
8. [Lay03, pp. 17–20] (Gaussian and Gauss–Jordan Elimination Algorithms) When one or more pivots are relatively small, using the algorithms below in floating point arithmetic can yield inaccurate results. (See Chapter 38 for more accurate variations of them, and Chapter 75 for information on professional software implementations of such variations.)

Algorithm 1. Gaussian and Gauss-Jordan Elimination

Let $A \in F^{m\times p}$. Steps 1 to 4 below do Gaussian Elimination, reducing A to a matrix that is in row echelon form. Steps 1 to 6 do Gauss–Jordan Elimination, reducing A to RREF(A).

1. Let $U = A$ and $r = 1$. If $U = \mathbf{0}$, U is in RREF.
2. If $U \neq \mathbf{0}$, search the submatrix of U in rows r to m to find its first nonzero column, k, and the first nonzero entry, a_{ik}, in this column. If $i > r$, exchange rows r and i in U, thus getting a nonzero entry in position (r, k). Let U be the matrix created by this row exchange.
3. Add multiples of row r to the rows below it, to create zeros in column k below row r. Let U denote the new matrix.
4. If either $r = m - 1$ or rows $r + 1, \dots, m$ are all zero, U is now in REF. Otherwise, let $r = r + 1$ and repeat steps 2, 3, and 4.
5. Let $k_1, \dots, k_s$ be the pivot columns of U, so $(1, k_1), \dots, (s, k_s)$ are the pivot positions. For $i = s$, $s - 1, \dots, 2$, add multiples of row i to the rows above it to create zeros in column k_i above row i.
6. For $i = 1, \dots, s$, divide row s by its leading entry. The resulting matrix is RREF(A).

Examples:

1. The RREF of a zero matrix is itself, and its rank is zero.
2. Let $A = \begin{bmatrix} 1 & 3 & 4 & -8 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 4 & -8 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. Both are upper triangular, but A is in REF and B is not. Use Gauss–Jordan Elimination to calculate RREF(A) and RREF(B).

   For A, add (-2)(row 2) to row 1 and multiply row 2 by $\frac{1}{2}$. This yields RREF$(A) = \begin{bmatrix} 1 & 3 & 0 & -16 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

   For B, exchange rows 2 and 3 to get $\begin{bmatrix} 1 & 3 & 4 & -8 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$, which is in REF. Then add 2(row 3) to row 1 to get a new matrix. In this new matrix, add (-4)(row 2) to row 1, and multiply row 3 by $\frac{1}{4}$. This yields RREF$(B) = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

   Observe that rank $(A) = 2$ and rank $(B) = 3$.
3. Apply Gauss–Jordan Elimination to $A = \begin{bmatrix} 2 & 6 & 4 & 4 \\ -4 & -12 & -8 & -7 \\ 0 & 0 & -1 & -4 \\ 1 & 3 & 1 & -2 \end{bmatrix}$.

   Step 1. Let $U^{(1)} = A$ and $r = 1$.
   Step 2. No row exchange is needed since $a_{11} \neq 0$.

   Step 3. Add (2)(row 1) to row 2, and $(-\frac{1}{2})$(row 1) to row 4 to get $U^{(2)} = \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -1 & -4 \end{bmatrix}$.

   Step 4. The submatrix in rows 2, 3, 4 is not zero, so let $r = 2$ and return to Step 2.

Step 2. Search the submatrix in rows 2 to 4 of $U^{(2)}$ to see that its first nonzero column is column 3 and the first nonzero entry in this column is in row 3 of $U^{(2)}$. Exchange rows 2 and 3 in $U^{(2)}$ to get

$$U^{(3)} = \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -4 \end{bmatrix}.$$

Step 3. Add row 2 to row 4 in $U^{(3)}$ to get $U^{(4)} = \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Step 4. Now $U^{(4)}$ is in REF, so Gaussian Elimination is finished.

Step 5. The pivot positions are $(1, 1)$, $(2, 3)$, and $(3, 4)$. Add –4(row 3) to rows 1 and 2 of $U^{(4)}$ to get

$U^{(5)} = \begin{bmatrix} 2 & 6 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Add –4(row 2) of $U^{(5)}$ to row 1 of $U^{(5)}$ to get $U^{(6)} = \begin{bmatrix} 2 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Step 6. Multiply row 1 of $U^{(6)}$ by $\frac{1}{2}$, obtaining $U^{(7)} = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, which is RREF$(A)$.

1.4 Systems of Linear Equations

Definitions:

A **linear equation** is an equation of the form $a_1x_1 + \cdots + a_px_p = b$ where $a_1, \ldots, a_p, b \in F$ and $x_1, \ldots, x_p$ are **variables**. The scalars a_j are **coefficients** and the scalar b is the **constant term.**

A **system of linear equations**, or **linear system**, is a set of one or more linear equations in the same variables, such as $\begin{matrix} a_{11}x_1 + \cdots + a_{1p}x_p = b_1 \\ a_{21}x_2 + \cdots + a_{2p}x_p = b_2 \\ \cdots \\ a_{m1}x_1 + \cdots + a_{mp}x_p = b_m \end{matrix}$. A **solution** of the system is a p-tuple $(c_1, \ldots, c_p)$ such that letting $x_j = c_j$ for each j satisfies every equation. The **solution set** of the system is the set of all solutions. A system is **consistent** if there exists at least one solution; otherwise it is **inconsistent**. Systems are **equivalent** if they have the same solution set. If $b_j = 0$ for all j, the system is **homogeneous**. A formula that describes a general vector in the solution set is called the **general solution**.

For the system $\begin{matrix} a_{11}x_1 + \cdots + a_{1p}x_p = b_1 \\ a_{21}x_2 + \cdots + a_{2p}x_p = b_2 \\ \cdots \\ a_{m1}x_1 + \cdots + a_{mp}x_p = b_m \end{matrix}$, the $m \times p$ matrix $A = \begin{bmatrix} a_{11} & \cdots & a_{1p} \\ \vdots & \cdots & \vdots \\ a_{m1} & \cdots & a_{mp} \end{bmatrix}$ is the **coefficient matrix**, $\mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$ is the **constant vector**, and $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}$ is the **unknown vector**. The $m \times (p+1)$ matrix $[A\ \mathbf{b}]$ is the **augmented matrix** of the system. It is customary to identify the system of linear equations with the matrix-vector equation $A\mathbf{x} = \mathbf{b}$. This is valid because a column vector $\mathbf{x} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$ satisfies $A\mathbf{x} = \mathbf{b}$ if and only if $(c_1, \ldots, c_p)$ is a solution of the linear system.

Observe that the coefficients of x_k are stored in column k of A. If $A\mathbf{x} = \mathbf{b}$ is equivalent to $C\mathbf{x} = \mathbf{d}$ and column k of C is a pivot column, then x_k is a **basic variable**; otherwise, x_k is a **free variable**.

Facts:

Let $A\mathbf{x} = \mathbf{b}$ be a linear system, where A is an $m \times p$ matrix.

1. [SIF00, pp. 27, 118] If elementary row operations are done to the augmented matrix $[A\,\mathbf{b}]$, obtaining a new matrix $[C\ \mathbf{d}]$, the new system $C\mathbf{x} = \mathbf{d}$ is equivalent to $A\mathbf{x} = \mathbf{b}$.
2. [SIF00, p. 24] There are three possibilities for the solution set of $A\mathbf{x} = \mathbf{b}$: either there are no solutions or there is exactly one solution or there is more than one solution. If there is more than one solution and F is infinite (such as the real numbers or complex numbers), then there are infinitely many solutions. If there is more than one solution and F is finite, then there are at least $|F|$ solutions.
3. A homogeneous system is always consistent (the zero vector $\mathbf{0}$ is always a solution).
4. The set of solutions to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is a subspace of the vector space F^p.
5. [SIF00, p. 44] The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is not a pivot column of $[A\ \mathbf{b}]$, that is, if and only if $\operatorname{rank}([A\ \mathbf{b}]) = \operatorname{rank} A$.
6. [SIF00, pp. 29–32] Suppose $A\mathbf{x} = \mathbf{b}$ is consistent. It has a unique solution if and only there is a pivot position in each column of A, that is, if and only if there are no free variables in the equation $A\mathbf{x} = \mathbf{b}$. Suppose there are $t \geq 1$ nonpivot columns in A. Then there are t free variables in the system. If $\operatorname{RREF}([A\ \mathbf{b}]) = [C\ \mathbf{d}]$, then the general solution of $C\mathbf{x} = \mathbf{d}$, hence of $A\mathbf{x} = \mathbf{b}$, can be written in the form $\mathbf{x} = s_1\mathbf{v}_1 + \cdots + s_t\mathbf{v}_t + \mathbf{w}$ where $\mathbf{v}_1, \ldots, \mathbf{v}_t, \mathbf{w}$ are column vectors and $s_1, \ldots, s_t$ are parameters, each representing one of the free variables. Thus $\mathbf{x} = \mathbf{w}$ is one solution of $A\mathbf{x} = \mathbf{b}$. Also, the general solution of $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = s_1\mathbf{v}_1 + \cdots + s_t\mathbf{v}_t$.
7. [SIF00, pp. 29–32] (General solution of a linear system algorithm)

Algorithm 2: General Solution of a Linear System $A\mathbf{x} = \mathbf{b}$

This algorithm is intended for small systems using rational arithmetic. It is not the most efficient and when some pivots are relatively small, using this algorithm in floating point arithmetic can yield inaccurate results. (For more accurate and efficient algorithms, see Chapter 38.) Let $A \in F^{m\times p}$ and $\mathbf{b} \in F^{p\times 1}$.

1. Calculate $\operatorname{RREF}([A\ \mathbf{b}])$, obtaining $[C\ \mathbf{d}]$.
2. If there is a pivot in the last column of $[C\ \mathbf{d}]$, stop. There is no solution.
3. Assume the last column of $[C\ \mathbf{d}]$ is not a pivot column, and let $\mathbf{d} = [d_1, \ldots, d_m]^T$.
   a. If $\operatorname{rank}(C) = p$, so there exists a pivot in each column of C, then $\mathbf{x} = \mathbf{d}$ is the unique solution of the system.
   b. Suppose $\operatorname{rank} C = r < p$.
      i. Write the system of linear equations represented by the nonzero rows of $[C\ \mathbf{d}]$. In each equation, the first nonzero term will be a basic variable, and each basic variable appears in only one of these equations.
      ii. Solve each equation for its basic variable and substitute parameter names for the $p - r$ free variables, say $s_1, \ldots, s_{p-r}$. This is the general solution of $C\mathbf{x} = \mathbf{d}$ and, thus, the general solution of $A\mathbf{x} = \mathbf{b}$.
      iii. To write the general solution in vector form, as $\mathbf{x} = s_1\mathbf{v}^{(1)} + \cdots + s_{p-r}\mathbf{v}^{(p-r)} + \mathbf{w}$, let (i, k_i) be the i^{th} pivot position of C. Define $\mathbf{w} \in F^p$ by $w_{k_i} = d_i$ for $i = 1, \ldots, r$, and all other entries of $\mathbf{w}$ are 0. Let x_{u_j} be the j^{th} free variable, and define the vectors $\mathbf{v}^{(j)} \in F^p$ as follows:

         For $j = 1, \ldots, p - r$,
            the u_j-entry of $\mathbf{v}^{(j)}$ is 1,
            for $i = 1, \ldots, r$, the k_i-entry of $\mathbf{v}^{(j)}$ is $-c_{iu_j}$,
            and all other entries of $\mathbf{v}^{(j)}$ are 0.

Examples:

1. The linear system $\begin{aligned} x_1 + x_2 &= 0 \\ -x_1 + x_2 &= 0 \end{aligned}$ has augmented matrix $\begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \end{bmatrix}$. The RREF of this is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, which is the augmented matrix for the equivalent system $\begin{aligned} x_1 &= 0 \\ x_2 &= 0 \end{aligned}$. Thus, the original system has a unique solution in $\mathbb{R}^2$, (0,0). In vector form the solution is $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

2. The system $\begin{aligned} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 0 \end{aligned}$ has a unique solution in $\mathbb{R}^2$, $(1, 1)$, or $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

3. The system $\begin{aligned} x_1 + x_2 + x_3 &= 2 \\ x_2 + x_3 &= 2 \\ x_3 &= 0 \end{aligned}$ has a unique solution in $\mathbb{R}^3$, $(0, 2, 0)$, or $\mathbf{x} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$.

4. The system $\begin{aligned} x_1 + x_2 &= 2 \\ 2x_1 + 2x_2 &= 4 \end{aligned}$ has infinitely many solutions in $\mathbb{R}^2$. The augmented matrix reduces to $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$, so the only equation left is $x_1 + x_2 = 2$. Thus x_1 is basic and x_2 is free. Solving for x_1 and letting $x_2 = s$ gives $x_1 = -s + 2$. Then the general solution is $\begin{aligned} x_1 &= -s + 2 \\ x_2 &= s \end{aligned}$, or all vectors of the form $(-s + 2, s)$. Letting $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, the vector form of the general solution is $\mathbf{x} = \begin{bmatrix} -s + 2 \\ s \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix}$.

5. The system $\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 1 \\ x_2 + x_3 - x_4 &= 3 \end{aligned}$ has infinitely many solutions in $\mathbb{R}^4$. Its augmented matrix $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & -1 & 3 \end{bmatrix}$ reduces to $\begin{bmatrix} 1 & 0 & 0 & 2 & -2 \\ 0 & 1 & 1 & -1 & 3 \end{bmatrix}$. Thus, x_1 and x_2 are the basic variables, and x_3 and x_4 are free. Write each of the new equations and solve it for its basic variable to see $\begin{aligned} x_1 &= -2x_4 - 2 \\ x_2 &= -x_3 + x_4 + 3 \end{aligned}$. Let $x_3 = s_1$ and $x_4 = s_2$ to get the general solution $\begin{aligned} x_1 &= -2s_2 - 2 \\ x_2 &= -s_1 + s_2 + 3 \\ x_3 &= s_1 \\ x_4 &= s_2 \end{aligned}$, or $\mathbf{x} = s_1\mathbf{v}^{(1)} + s_2\mathbf{v}^{(2)} + \mathbf{w} = s_1 \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 3 \\ 0 \\ 0 \end{bmatrix}$.

6. These systems have no solutions: $\begin{aligned} x_1 + x_2 &= 0 \\ x_1 + x_2 &= 1 \end{aligned}$ and $\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ x_1 - x_2 - x_3 &= 0 \\ x_2 + x_3 &= 1 \end{aligned}$. This can be verified by inspection, or by calculating the RREF of the augmented matrix of each and observing that each has a pivot in its last column.

1.5 Matrix Inverses and Elementary Matrices

Invertibility is a strong and useful property. For example, when a linear system $A\mathbf{x} = \mathbf{b}$ has an invertible coefficient matrix A, it has a unique solution. The various characterizations of invertibility in Fact 10 below are also quite useful. Throughout this section, F will denote a field.

Definitions:

An $n \times n$ matrix A is **invertible**, or **nonsingular**, if there exists another $n \times n$ matrix B, called the **inverse** of A, such that $AB = BA = I_n$. The inverse of A is denoted A^{-1} (cf. Fact 1). If no such B exists, A is **not invertible**, or **singular**.

For an $n \times n$ matrix and a positive integer m, the ***m*th power** of A is $A^m = \underbrace{AA \dots A}_{m \text{ copies of } A}$. It is also convenient to define $A^0 = I_n$. If A is invertible, then $A^{-m} = (A^{-1})^m$.

An **elementary matrix** is a square matrix obtained by doing one elementary row operation to an identity matrix. Thus, there are three types:

1. A multiple of one row of I_n has been added to a different row.
2. Two different rows of I_n have been exchanged.
3. One row of I_n has been multiplied by a nonzero scalar.

Facts:

1. [SIF00, pp. 114–116] If $A \in F^{n\times n}$ is invertible, then its inverse is unique.
2. [SIF00, p. 128] (Method to compute A^{-1}) Suppose $A \in F^{n\times n}$. Create the matrix $[A\ I_n]$ and calculate its RREF, which will be of the form $[\text{RREF}(A)X]$. If $\text{RREF}(A) = I_n$, then A is invertible and $X = A^{-1}$. If $\text{RREF}(A) \neq I_n$, then A is not invertible. As with the Gaussian algorithm, this method is theoretically correct, but more accurate and efficient methods for calculating inverses are used in professional computer software. (See Chapter 75.)
3. [SIF00, pp. 114–116] If $A \in F^{n\times n}$ is invertible, then A^{-1} is invertible and $(A^{-1})^{-1} = A$.
4. [SIF00, pp. 114–116] If $A, B \in F^{n\times n}$ are invertible, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
5. [SIF00, pp. 114–116] If $A \in F^{n\times n}$ is invertible, then A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.
6. If $A \in F^{n\times n}$ is invertible, then for each $\mathbf{b} \in F^{n\times 1}$, $A\mathbf{x} = \mathbf{b}$ has a unique solution, and it is $\mathbf{x} = A^{-1}\mathbf{b}$.
7. [SIF00, p. 124] If $A \in F^{n\times n}$ and there exists $C \in F^{n\times n}$ such that either $AC = I_n$ or $CA = I_n$, then A is invertible and $A^{-1} = C$. That is, a left or right inverse for a square matrix is actually its unique two-sided inverse.
8. [SIF00, p. 117] Let E be an elementary matrix obtained by doing one elementary row operation to I_n. If that same row operation is done to an $n \times p$ matrix A, the result equals EA.
9. [SIF00, p. 117] An elementary matrix is invertible and its inverse is another elementary matrix of the same type.
10. [SIF00, pp. 126] (Invertible Matrix Theorem) (See Section 2.5.) When $A \in F^{n\times n}$, the following are equivalent:
    - A is invertible.
    - $\text{RREF}(A) = I_n$.
    - $\text{Rank}(A) = n$.
    - The only solution of $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
    - For every $\mathbf{b} \in F^{n\times 1}$, $A\mathbf{x} = \mathbf{b}$ has a unique solution.
    - For every $\mathbf{b} \in F^{n\times 1}$, $A\mathbf{x} = \mathbf{b}$ has a solution.
    - There exists $B \in F^{n\times n}$ such that $AB = I_n$.
    - There exists $C \in F^{n\times n}$ such that $CA = I_n$.
    - A^T is invertible.
    - There exist elementary matrices whose product equals A.
11. [SIF00, p. 148] and [Lay03, p.132] Let $A \in F^{n\times n}$ be upper (lower) triangular. Then A is invertible if and only if each diagonal entry is nonzero. If A is invertible, then A^{-1} is also upper (lower) triangular, and the diagonal entries of A^{-1} are the reciprocals of those of A. In particular, if L is a unit upper (lower) triangular matrix, then L^{-1} is also a unit upper (lower) triangular matrix.

12. Matrix powers obey the usual rules of exponents, i.e., when A^s and A^t are defined for integers s and t, then $A^sA^t = A^{s+t}, (A^s)^t = A^{st}$.

Examples:

1. For any n, the identity matrix I_n is invertible and is its own inverse. If P is a permutation matrix, it is invertible and $P^{-1} = P^T$.
2. If $A = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -3 \\ -2 & 7 \end{bmatrix}$, then calculation shows $AB = BA = I_2$, so A is invertible and $A^{-1} = B$.
3. If $A = \begin{bmatrix} 0.2 & 4 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$, then $A^{-1} = \begin{bmatrix} 5 & -10 & -5 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & -1 \end{bmatrix}$, as can be verified by multiplication.
4. The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is not invertible since $\text{RREF}(A) \neq I_2$. Alternatively, if B is any 2×2 matrix, AB is of the form $\begin{bmatrix} r & s \\ 2r & 2s \end{bmatrix}$, which cannot equal I_2.
5. Let A be an $n \times n$ matrix A with a zero row (zero column). Then A is not invertible since $\text{RREF}(A) \neq I_n$. Alternatively, if B is any $n \times n$ matrix, AB has a zero row (BA has a zero column), so B is not an inverse for A.
6. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is any 2×2 matrix, then A is invertible if and only if $ad - bc \neq 0$; further, when $ad - bc \neq 0$, $A^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. The scalar $ad - bc$ is called the determinant of A. (The determinant is defined for any $n \times n$ matrix in Section 4.1.) Using this formula, the matrix $A = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix}$ from Example 2 (above) has determinant 1, so A is invertible and $A^{-1} = \begin{bmatrix} 1 & -3 \\ -2 & 7 \end{bmatrix}$, as noted above. The matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ from Example 3 (above) is not invertible since its determinant is 0.
7. Let $A = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 7 & 0 \\ 1 & 1 & 1 \end{bmatrix}$. Then $\text{RREF}([A \quad I_n]) = \begin{bmatrix} 1 & 0 & 0 & 7 & -3 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -5 & 2 & 1 \end{bmatrix}$, so A^{-1} exists and equals $\begin{bmatrix} 7 & -3 & 0 \\ -2 & 1 & 0 \\ -5 & 2 & 1 \end{bmatrix}$.

1.6 LU Factorization

This section discusses the *LU* and *PLU* factorizations of a matrix that arise naturally when Gaussian Elimination is done. Several other factorizations are widely used for real and complex matrices, such as the QR, Singular Value, and Cholesky Factorizations. (See Chapter 5 and Chapter 38.) Throughout this section, F will denote a field and A will denote a matrix over F. The material in this section and additional background can be found in [GV96, Sec. 3.2].

Definitions:

Let A be a matrix of any shape.

An LU **factorization**, or **triangular factorization**, of A is a factorization $A = LU$ where L is a square unit lower triangular matrix and U is upper triangular. A ***PLU*** **factorization** of A is a factorization of the form $PA = LU$ where P is a permutation matrix, L is square unit lower triangular, and U is upper triangular. An ***LDU* factorization** of A is a factorization $A = LDU$ where L is a square unit lower triangular matrix, D is a square diagonal matrix, and U is a unit upper triangular matrix.

A ***PLDU* factorization** of A is a factorization $PA = LDU$ where P is a permutation matrix, L is a square unit lower triangular matrix, D is a square diagonal matrix, and U is a unit upper triangular matrix.

Facts: [GV96, Sec. 3.2]

1. Let A be square. If each leading principal submatrix of A, except possibly A itself, is invertible, then A has an LU factorization. When A is invertible, A has an LU factorization if and only if each leading principal submatrix of A is invertible; in this case, the LU factorization is unique and there is also a unique LDU factorization of A.
2. Any matrix A has a PLU factorization. Algorithm 1 (Section 1.3) performs the addition of multiples of pivot rows to lower rows and perhaps row exchanges to obtain an REF matrix U. If instead, the same series of row exchanges are done to A before any pivoting, this creates PA where P is a permutation matrix, and then PA can be reduced to U without row exchanges. That is, there exist unit lower triangular matrices E_j such that $E_k \dots E_1(PA) = U$. It follows that $PA = LU$, where $L = (E_k \dots E_1)^{-1}$ is unit lower triangular and U is upper triangular.
3. In most professional software packages, the standard method for solving a square linear system $A\mathbf{x} = \mathbf{b}$, for which A is invertible, is to reduce A to an REF matrix U as in Fact 2 above, choosing row exchanges by a strategy to reduce pivot size. By keeping track of the exchanges and pivot operations done, this produces a PLU factorization of A. Then $A = P^T LU$ and $P^T LU\mathbf{x} = \mathbf{b}$ is the equation to be solved. Using forward substitution, $P^T L\mathbf{y} = \mathbf{b}$ can be solved quickly for $\mathbf{y}$, and then $U\mathbf{x} = \mathbf{y}$ can either be solved quickly for $\mathbf{x}$ by back substitutution, or be seen to be inconsistent. This method gives accurate results for most problems. There are other types of solution methods that can work more accurately or efficiently for special types of matrices. (See Chapter 7.)

Examples:

1. Calculate a PLU factorization for $A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ -1 & -1 & -3 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & -1 & 1 \end{bmatrix}$. If Gaussian Elimination is performed on A, after adding row 1 to rows 2 and 4, rows 2 and 3 must be exchanged and the final result is $U = E_3 P E_2 E_1 A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ where E_1, E_2, and E_3 are lower triangular unit matrices and P is a permutation matrix. This will not yield an LU factorization of A. But if the row exchange is done to A first, by multiplying A by $P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, one gets $PA = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ -1 & -1 & -3 & 1 \\ -1 & 0 & -1 & 1 \end{bmatrix}$; then Gaussian Elimination can proceed without any row exchanges. Add row 1 to rows 3 and 4 to get $F_2 F_1 PA = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 4 \\ 0 & 1 & 1 & 4 \end{bmatrix}$ where $F_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ and $F_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$. Then add $(-1)(\text{row } 2)$ to row 4 to get $U = F_3\, F_2\, F_1\, PA = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix}$, where $F_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$.

Note that U is the same upper triangular matrix as before. Finally, $L = (F_3\, F_2\, F_1)^{-1}$ is unit lower triangular and $PA = LU$ is true, so this is a *PLU* factorization of A. To get a *PLDU* factorization, use the same P and L, and define $D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

2. Let $A = LU = \begin{bmatrix} 1 & 3 & 4 \\ -1 & -1 & -5 \\ 2 & 12 & 3 \end{bmatrix}$. Each leading principal submatrix of A is invertible so A has both *LU* and *LDU* factorizations:
$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 \\ 0 & 2 & -1 \\ 0 & 0 & -2 \end{bmatrix}$. This yields an *LDU* factorization of A, $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ $\begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & -0.5 \\ 0 & 0 & 1 \end{bmatrix}$. With the *LU* factorization, an equation such as $A\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ can be solved efficiently as follows. Use forward substitution to solve $L\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, getting $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ -8 \end{bmatrix}$, and then backward substitution to solve $U\mathbf{x} = \mathbf{y}$, getting $\mathbf{x} = \begin{bmatrix} -24 \\ 3 \\ 4 \end{bmatrix}$.
3. Any invertible matrix whose $(1, 1)$ entry is zero, such as $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ or $\begin{bmatrix} 0 & -1 & 5 \\ 1 & 1 & 1 \\ 1 & 0 & 3 \end{bmatrix}$, does not have an *LU* factorization.
4. The matrix $A = \begin{bmatrix} 1 & 3 & 4 \\ -1 & -3 & -5 \\ 2 & 6 & 6 \end{bmatrix}$ is not invertible, nor is its leading principal 2×2 submatrix, but it does have an *LU* factorization: $A = LU = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$. To find out if an equation such as $A\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is consistent, notice $L\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ yields $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ -8 \end{bmatrix}$, but $U\mathbf{x} = \mathbf{y}$ is inconsistent, hence $A\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ has no solution.
5. The matrix $A = \begin{bmatrix} 0 & -1 & 5 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ has no *LU* factorization, but does have a *PLU* factorization with $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$, and $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 5 \\ 0 & 0 & -4 \end{bmatrix}$.

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2

Linear Independence, Span, and Bases

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2.1 Span and Linear Independence

Let V be a vector space over a field F.

Definitions:

A **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in V$ is a sum of scalar multiples of these vectors; that is, $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$, for some scalar coefficients $c_1, c_2, \ldots, c_k \in F$. If S is a set of vectors in V, a linear combination of vectors in S is a vector of the form $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$ with $k \in \mathbb{N}, \mathbf{v}_i \in S, c_i \in F$. Note that S may be finite or infinite, but a linear combination is, by definition, a finite sum. The zero vector is defined to be a linear combination of the empty set.

When all the scalar coefficients in a linear combination are 0, it is a **trivial linear combination.** A sum over the empty set is also a trivial linear combination.

The **span** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in V$ is the set of all linear combinations of these vectors, denoted by $\mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$. If S is a (finite or infinite) set of vectors in V, then the span of S, denoted by $\mathrm{Span}(S)$, is the set of all linear combinations of vectors in S.

If $V = \mathrm{Span}(S)$, then S **spans** the vector space V.

A (finite or infinite) set of vectors S in V is **linearly independent** if the only linear combination of distinct vectors in S that produces the zero vector is a trivial linear combination. That is, if $\mathbf{v}_i$ are distinct vectors in S and $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$, then $c_1 = c_2 = \cdots = c_k = 0$. Vectors that are not linearly independent are **linearly dependent**. That is, there exist distinct vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in S$ and $c_1, c_2, \ldots, c_k$ not all 0 such that $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$.

Facts: The following facts can be found in [Lay03, Sections 4.1 and 4.3].

1. $\mathrm{Span}(\emptyset) = \{\mathbf{0}\}$.
2. A linear combination of a single vector $\mathbf{v}$ is simply a scalar multiple of $\mathbf{v}$.
3. In a vector space V, $\mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ is a subspace of V.
4. Suppose the set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ spans the vector space V. If one of the vectors, say $\mathbf{v}_i$, is a linear combination of the remaining vectors, then the set formed from S by removing $\mathbf{v}_i$ still spans V.
5. Any single nonzero vector is linearly independent.
6. Two nonzero vectors are linearly independent if and only if neither is a scalar multiple of the other.
7. If S spans V and $S \subseteq T$, then T spans V.
8. If T is a linearly independent subset of V and $S \subseteq T$, then S is linearly independent.
9. Vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ are linearly dependent if and only if $\mathbf{v}_i = c_1\mathbf{v}_1 + \cdots + c_{i-1}\mathbf{v}_{i-1} + c_{i+1}\mathbf{v}_{i+1} + \cdots + c_k\mathbf{v}_k$, for some $1 \leq i \leq k$ and some scalars $c_1, \ldots, c_{i-1}, c_{i+1}, \ldots, c_k$. A set S of vectors in V is linearly dependent if and only if there exists $\mathbf{v} \in S$ such that $\mathbf{v}$ is a linear combination of other vectors in S.
10. Any set of vectors that includes the zero vector is linearly dependent.

Examples:

1. Linear combinations of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix} \in \mathbb{R}^2$ are vectors of the form $c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} c_1 \\ -c_1 + 3c_2 \end{bmatrix}$, for any scalars $c_1, c_2 \in \mathbb{R}$. Any vector of this form is in $\mathrm{Span}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix}\right)$. In fact, $\mathrm{Span}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix}\right) = \mathbb{R}^2$ and these vectors are linearly independent.
2. If $\mathbf{v} \in \mathbb{R}^n$ and $\mathbf{v} \neq \mathbf{0}$, then geometrically $\mathrm{Span}(\mathbf{v})$ is a line in $\mathbb{R}^n$ through the origin.
3. Suppose $n \geq 2$ and $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$ are linearly independent vectors. Then geometrically $\mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2)$ is a plane in $\mathbb{R}^n$ through the origin.
4. Any polynomial $p(x) \in \mathbb{R}[x]$ of degree less than or equal to 2 can easily be seen to be a linear combination of $1, x$, and x^2. However, $p(x)$ is also a linear combination of $1, 1+x$, and $1+x^2$. So $\mathrm{Span}(1, x, x^2) = \mathrm{Span}(1, 1+x, 1+x^2) = \mathbb{R}[x; 2]$.
5. The n vectors $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \ldots, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$ span F^n, for any field F. These vectors are also linearly independent.
6. In $\mathbb{R}^2$, $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$ are linearly independent. However, $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 5 \end{bmatrix}$ are linearly dependent, because $\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 3 \end{bmatrix}$.
7. The infinite set $\{1, x, x^2, \ldots, x^n, \ldots\}$ is linearly independent in $F[x]$, for any field F.
8. In the vector space of continuous real-valued functions on the real line, $\mathcal{C}(\mathbb{R})$, the set $\{\sin(x), \sin(2x), \ldots, \sin(nx), \cos(x), \cos(2x), \ldots, \cos(nx)\}$ is linearly independent for any $n \in \mathbb{N}$. The infinite set $\{\sin(x), \sin(2x), \ldots, \sin(nx), \ldots, \cos(x), \cos(2x), \ldots, \cos(nx), \ldots\}$ is also linearly independent in $\mathcal{C}(\mathbb{R})$.

Applications:

1. The homogeneous differential equation $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = 0$ has as solutions $y_1(x) = e^{2x}$ and $y_2(x) = e^x$. Any linear combination $y(x) = c_1y_1(x) + c_2y_2(x)$ is a solution of the differential equation, and so $\text{Span}(e^{2x}, e^x)$ is contained in the set of solutions of the differential equation (called the solution space for the differential equation). In fact, the solution space is spanned by e^{2x} and e^x, and so is a subspace of the vector space of functions. In general, the solution space for a homogeneous differential equation is a vector space, meaning that any linear combination of solutions is again a solution.

2.2 Basis and Dimension of a Vector Space

Let V be a vector space over a field F.

Definitions:

A set of vectors $\mathcal{B}$ in a vector space V is a **basis** for V if

- $\mathcal{B}$ is a linearly independent set, and
- $\text{Span}(\mathcal{B}) = V$.

The set $\mathcal{E}_n = \left\{ \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \ldots, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \right\}$ is the **standard basis** for F^n.

The number of vectors in a basis for a vector space V is the **dimension** of V, denoted by $\dim(V)$. If a basis for V contains a finite number of vectors, then V is **finite dimensional**. Otherwise, V is **infinite dimensional**, and we write $\dim(V) = \infty$.

Facts: All the following facts, except those with a specific reference, can be found in [Lay03, Sections 4.3 and 4.5].

1. Every vector space has a basis.
2. The standard basis for F^n is a basis for F^n, and so $\dim F^n = n$.
3. A basis $\mathcal{B}$ in a vector space V is the largest set of linearly independent vectors in V that contains $\mathcal{B}$, and it is the smallest set of vectors in V that contains $\mathcal{B}$ and spans V.
4. The empty set is a basis for the trivial vector space $\{\mathbf{0}\}$, and $\dim(\{\mathbf{0}\}) = 0$.
5. If the set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ spans a vector space V, then some subset of S forms a basis for V. In particular, if one of the vectors, say $\mathbf{v}_i$, is a linear combination of the remaining vectors, then the set formed from S by removing $\mathbf{v}_i$ will be "closer" to a basis for V. This process can be continued until the remaining vectors form a basis for V.
6. If S is a linearly independent set in a vector space V, then S can be expanded, if necessary, to a basis for V.
7. No nontrivial vector space over a field with more than two elements has a unique basis.
8. If a vector space V has a basis containing n vectors, then every basis of V must contain n vectors. Similarly, if V has an infinite basis, then every basis of V must be infinite. So the dimension of V is unique.
9. Let $\dim(V) = n$ and let S be a set containing n vectors. The following are equivalent:
   - S is a basis for V.
   - S spans V.
   - S is linearly independent.

10. If $\dim(V) = n$, then any subset of V containing more than n vectors is linearly dependent.
11. If $\dim(V) = n$, then any subset of V containing fewer than n vectors does not span V.
12. [Lay03, Section 4.4] If $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$ is a basis for a vector space V, then each $\mathbf{x} \in V$ can be expressed as a unique linear combination of the vectors in $\mathcal{B}$. That is, for each $\mathbf{x} \in V$ there is a unique set of scalars $c_1, c_2, \ldots, c_p$ such that $\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_p\mathbf{b}_p$.

Examples:

1. In $\mathbb{R}^2$, $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$ are linearly independent, and they span $\mathbb{R}^2$. So they form a basis for $\mathbb{R}^2$ and $\dim(\mathbb{R}^2) = 2$.
2. In $F[x]$, the set $\{1, x, x^2, \ldots, x^n\}$ is a basis for $F[x; n]$ for any $n \in \mathbb{N}$. The infinite set $\{1, x, x^2, x^3, \ldots\}$ is a basis for $F[x]$, meaning $\dim(F[x]) = \infty$.
3. The set of $m \times n$ matrices E_{ij} having a 1 in the i, j-entry and zeros everywhere else forms a basis for $F^{m\times n}$. Since there are mn such matrices, $\dim(F^{m\times n}) = mn$.
4. The set $S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ clearly spans $\mathbb{R}^2$, but it is not a linearly independent set. However, removing any single vector from S will cause the remaining vectors to be a basis for $\mathbb{R}^2$, because any pair of vectors is linearly independent and still spans $\mathbb{R}^2$.
5. The set $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ is linearly independent, but it cannot be a basis for $\mathbb{R}^4$ since it does not span $\mathbb{R}^4$. However, we can start expanding it to a basis for $\mathbb{R}^4$ by first adding a vector that is not in the span of S, such as $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$. Then since these three vectors still do not span $\mathbb{R}^4$, we can add a vector that is not in their span, such as $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. These four vectors now span $\mathbb{R}^4$ and they are linearly independent, so they form a basis for $\mathbb{R}^4$.
6. Additional techniques for determining whether a given finite set of vectors is linearly independent or spans a given subspace can be found in Sections 2.5 and 2.6.

Applications:

1. Because $y_1(x) = e^{2x}$ and $y_2(x) = e^x$ are linearly independent and span the solution space for the homogeneous differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$, they form a basis for the solution space and the solution space has dimension 2.

2.3 Direct Sum Decompositions

Throughout this section, V will be a vector space over a field F, and W_i, for $i = 1, \ldots, k$, will be subspaces of V. For facts and general reading for this section, see [HK71].

Definitions:

The **sum** of subspaces W_i, for $i = 1, \dots, k$, is $\sum_{i=1}^{k} W_i = W_1 + \cdots + W_k = \{\mathbf{w}_1 + \cdots + \mathbf{w}_k \mid \mathbf{w}_i \in W_i\}$. The sum $W_1 + \cdots + W_k$ is a **direct sum** if for all $i = 1, \dots, k$, we have $W_i \cap \sum_{j \neq i} W_j = \{\mathbf{0}\}$. $W = W_1 \oplus \cdots \oplus W_k$ denotes that $W = W_1 + \cdots + W_k$ and the sum is direct. The subspaces W_i, for $i = i, \dots, k$, are **independent** if for $\mathbf{w}_i \in W_i$, $\mathbf{w}_1 + \cdots + \mathbf{w}_k = \mathbf{0}$ implies $\mathbf{w}_i = \mathbf{0}$ for all $i = 1, \dots, k$. Let V_i, for $i = 1, \dots, k$, be vector spaces over F. The **external direct sum** of the V_i, denoted $V_1 \times \cdots \times V_k$, is the cartesian product of V_i, for $i = 1, \dots, k$, with coordinate-wise operations. Let W be a subspace of V. An **additive coset** of W is a subset of the form $v + W = \{v + w \mid w \in W\}$ with $v \in V$. The **quotient** of V by W, denoted V/W, is the set of additive cosets of W with operations $(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$ and $c(v + W) = (cv) + W$, for any $c \in F$. Let $V = W \oplus U$, let $\mathcal{B}_W$ and $\mathcal{B}_U$ be bases for W and U respectively, and let $\mathcal{B} = \mathcal{B}_W \cup \mathcal{B}_U$. The **induced basis** of $\mathcal{B}$ in V/W is the set of vectors $\{u + W \mid u \in \mathcal{B}_U\}$.

Facts:

1. $W = W_1 \oplus W_2$ if and only if $W = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$.
2. If W is a subspace of V, then there exists a subspace U of V such that $V = W \oplus U$. Note that U is not usually unique.
3. Let $W = W_1 + \cdots + W_k$. The following are equivalent:
   - $W = W_1 \oplus \cdots \oplus W_k$. That is, for all $i = 1, \dots, k$, we have $W_i \cap \sum_{j \neq i} W_j = \{\mathbf{0}\}$.
   - $W_i \cap \sum_{j=1}^{i-1} W_j = \{\mathbf{0}\}$, for all $i = 2, \dots, k$.
   - For each $\mathbf{w} \in W$, $\mathbf{w}$ can be expressed in exactly one way as a sum of vectors in $W_1, \dots, W_k$. That is, there exist unique $\mathbf{w}_i \in W_i$, such that $\mathbf{w} = \mathbf{w}_1 + \cdots + \mathbf{w}_k$.
   - The subspaces W_i, for $i = 1, \dots, k$, are independent.
   - If $\mathcal{B}_i$ is an (ordered) basis for W_i, then $\mathcal{B} = \bigcup_{i=1}^{k} \mathcal{B}_i$ is an (ordered) basis for W.
4. If $\mathcal{B}$ is a basis for V and $\mathcal{B}$ is partitioned into disjoint subsets $\mathcal{B}_i$, for $i = 1, \dots, k$, then $V = \mathrm{Span}(\mathcal{B}_1) \oplus \cdots \oplus \mathrm{Span}(\mathcal{B}_k)$.
5. If S is a linearly independent subset of V and S is partitioned into disjoint subsets S_i, for $i = 1, \dots, k$, then the subspaces $\mathrm{Span}(S_1), \dots, \mathrm{Span}(S_k)$ are independent.
6. If V is finite dimensional and $V = W_1 + \cdots + W_k$, then $\dim(V) = \dim(W_1) + \cdots + \dim(W_k)$ if and only if $V = W_1 \oplus \cdots \oplus W_k$.
7. Let V_i, for $i = 1, \dots, k$, be vector spaces over F.
   - $V_1 \times \cdots \times V_k$ is a vector space over F.
   - $\widehat{V_i} = \{(0, \dots, 0, v_i, 0, \dots, 0) \mid v_i \in V_i\}$ (where v_i is the ith coordinate) is a subspace of $V_1 \times \cdots \times V_k$.
   - $V_1 \times \cdots \times V_k = \widehat{V_1} \oplus \cdots \oplus \widehat{V_k}$.
   - If V_i, for $i = 1, \dots, k$, are finite dimensional, then $\dim \widehat{V_i} = \dim V_i$ and $\dim(V_1 \times \cdots \times V_k) = \dim V_1 + \cdots + \dim V_k$.
8. If W is a subspace of V, then the quotient V/W is a vector space over F.
9. Let $V = W \oplus U$, let $\mathcal{B}_W$ and $\mathcal{B}_U$ be bases for W and U respectively, and let $\mathcal{B} = \mathcal{B}_W \cup \mathcal{B}_U$. The induced basis of $\mathcal{B}$ in V/W is a basis for V/W and $\dim(V/W) = \dim U$.

Examples:

1. Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V. Then $V = \mathrm{Span}(\mathbf{v}_1) \oplus \cdots \oplus \mathrm{Span}(\mathbf{v}_n)$.
2. Let $X = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} \mid x \in \mathbb{R} \right\}$, $Y = \left\{ \begin{bmatrix} 0 \\ y \end{bmatrix} \mid y \in \mathbb{R} \right\}$, and $Z = \left\{ \begin{bmatrix} z \\ z \end{bmatrix} \mid z \in \mathbb{R} \right\}$. Then $\mathbb{R}^2 = X \oplus Y = Y \oplus Z = X \oplus Z$.

3. In $F^{n\times n}$, let W_1 be the subspace of symmetric matrices and W_2 be the subspace of skew-symmetric matrices. Clearly, $W_1 \cap W_2 = \{\mathbf{0}\}$. For any $A \in F^{n\times n}$, $A = \frac{A+A^T}{2} + \frac{A-A^T}{2}$, where $\frac{A+A^T}{2} \in W_1$ and $\frac{A-A^T}{2} \in W_2$. Therefore, $F^{n\times n} = W_1 \oplus W_2$.
4. Recall that the function $f \in \mathcal{C}(\mathbb{R})$ is even if $f(-x) = f(x)$ for all x, and f is odd if $f(-x) = -f(x)$ for all x. Let W_1 be the subspace of even functions and W_2 be the subspace of odd functions. Clearly, $W_1 \cap W_2 = \{\mathbf{0}\}$. For any $f \in \mathcal{C}(\mathbb{R})$, $f = f_1 + f_2$, where $f_1(x) = \frac{f(x)+f(-x)}{2} \in W_1$ and $f_1(x) = \frac{f(x)-f(-x)}{2} \in W_2$. Therefore, $\mathcal{C}(\mathbb{R}) = W_1 \oplus W_2$.
5. Given a subspace W of V, we can find a subspace U such that $V = W \oplus U$ by choosing a basis for W, extending this linearly independent set to a basis for V, and setting U equal to the span of the basis vectors not in W. For example, in $\mathbb{R}^3$, Let $W = \left\{ \begin{bmatrix} a \\ -2a \\ a \end{bmatrix} \mid a \in \mathbb{R} \right\}$. If $\mathbf{w} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, then $\{\mathbf{w}\}$ is a basis for W. Extend this to a basis for $\mathbb{R}^3$, for example by adjoining $\mathbf{e}_1$ and $\mathbf{e}_2$. Thus, $V = W \oplus U$, where $U = \operatorname{Span}(\mathbf{e}_1, \mathbf{e}_2)$. Note: there are many other ways to extend the basis, and many other possible U.
6. In the external direct sum $\mathbb{R}[x;2] \times \mathbb{R}^{2\times 2}$, $\left(2x^2+7, \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right) + 3\left(x^2+4x-2, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right) = \left(5x^2+12x+1, \begin{bmatrix} 1 & 5 \\ 0 & 4 \end{bmatrix}\right)$.
7. The subspaces X, Y, Z of $\mathbb{R}^2$ in Example 2 have bases $\mathcal{B}_X = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$, $\mathcal{B}_Y = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$, $\mathcal{B}_Z = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$, respectively. Then $\mathcal{B}_{XY} = \mathcal{B}_X \cup \mathcal{B}_Y$ and $\mathcal{B}_{XZ} = \mathcal{B}_X \cup \mathcal{B}_Z$ are bases for $\mathbb{R}^2$. In $\mathbb{R}^2/X$, the induced bases of $\mathcal{B}_{XY}$ and $\mathcal{B}_{XZ}$ are $\left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} + X \right\}$ and $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} + X \right\}$, respectively. These are equal because $\begin{bmatrix} 1 \\ 1 \end{bmatrix} + X = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} + X = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + X$.

2.4 Matrix Range, Null Space, Rank, and the Dimension Theorem

Definitions:

For any matrix $A \in F^{m\times n}$, the **range** of A, denoted by range(A), is the set of all linear combinations of the columns of A. If $A = [\mathbf{m}_1\ \mathbf{m}_2\ \ldots\ \mathbf{m}_n]$, then range$(A) = \operatorname{Span}(\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_n)$. The range of A is also called the **column space** of A.

The **row space** of A, denoted by RS(A), is the set of all linear combinations of the rows of A. If $A = [\mathbf{v}_1\ \mathbf{v}_2\ \ldots\ \mathbf{v}_m]^T$, then RS$(A) = \operatorname{Span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m)$.

The **kernel** of A, denoted by ker(A), is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$. The kernel of A is also called the **null space** of A, and its dimension is called the **nullity** of A, denoted by null(A).

The **rank** of A, denoted by rank(A), is the number of leading entries in the reduced row echelon form of A (or any row echelon form of A). (See Section 1.3 for more information.)

$A, B \in F^{m\times n}$ are **equivalent** if $B = C_1^{-1}AC_2$ for some invertible matrices $C_1 \in F^{m\times m}$ and $C_2 \in F^{n\times n}$. $A, B \in F^{n\times n}$ are **similar** if $B = C^{-1}AC$ for some invertible matrix $C \in F^{n\times n}$. For square matrices $A_1 \in F^{n_1\times n_1}, \ldots, A_k \in F^{n_k\times n_k}$, the **matrix direct sum** $A = A_1 \oplus \cdots \oplus A_k$ is the block diagonal matrix with the matrices A_i down the diagonal. That is, $A = \begin{bmatrix} A_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & A_k \end{bmatrix}$, where $A \in F^{n\times n}$ with $n = \sum_{i=1}^{k} n_i$.

Facts: Unless specified otherwise, the following facts can be found in [Lay03, Sections 2.8, 4.2, 4.5, and 4.6].

1. The range of an $m \times n$ matrix A is a subspace of F^m.
2. The columns of A corresponding to the pivot columns in the reduced row echelon form of A (or any row echelon form of A) give a basis for range(A). Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in F^m$. If matrix $A = [\mathbf{v}_1\ \mathbf{v}_2\ \ldots\ \mathbf{v}_k]$, then a basis for range(A) will be a linearly independent subset of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ having the same span.
3. dim(range(A)) = rank(A).
4. The kernel of an $m \times n$ matrix A is a subspace of F^n.
5. If the reduced row echelon form of A (or any row echelon form of A) has k pivot columns, then null$(A) = n - k$.
6. If two matrices A and B are row equivalent, then RS(A) = RS(B).
7. The row space of an $m \times n$ matrix A is a subspace of F^n.
8. The pivot rows in the reduced row echelon form of A (or any row echelon form of A) give a basis for RS(A).
9. dim(RS(A)) = rank(A).
10. rank(A) = rank(A^T).
11. (Dimension Theorem) For any $A \in F^{m\times n}$, $n = \text{rank}(A) + \text{null}(A)$. Similarly, $m = \dim(\text{RS}(A)) + \text{null}(A^T)$.
12. A vector $\mathbf{b} \in F^m$ is in range(A) if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution. So range$(A) = F^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in F^m$.
13. A vector $\mathbf{a} \in F^n$ is in RS(A) if and only if the equation $A^T\mathbf{y} = \mathbf{a}$ has a solution. So RS$(A) = F^n$ if and only if the equation $A^T\mathbf{y} = \mathbf{a}$ has a solution for every $\mathbf{a} \in F^n$.
14. If $\mathbf{a}$ is a solution to the equation $A\mathbf{x} = \mathbf{b}$, then $\mathbf{a} + \mathbf{v}$ is also a solution for any $\mathbf{v} \in \ker(A)$.
15. [HJ85, p. 14] If $A \in F^{m\times n}$ is rank 1, then there are vectors $\mathbf{v} \in F^m$ and $\mathbf{u} \in F^n$ so that $A = \mathbf{v}\mathbf{u}^T$.
16. If $A \in F^{m\times n}$ is rank k, then A is a sum of k rank 1 matrices. That is, there exist $A_1, \ldots, A_k$ with $A = A_1 + \cdots + A_k$ and rank$(A_i) = 1$, for $i = 1, \ldots, k$.
17. [HJ85, p. 13] The following are all equivalent statements about a matrix $A \in F^{m\times n}$.
    (a) The rank of A is k.
    (b) dim(range(A)) $= k$.
    (c) The reduced row echelon form of A has k pivot columns.
    (d) A row echelon form of A has k pivot columns.
    (e) The largest number of linearly independent columns of A is k.
    (f) The largest number of linearly independent rows of A is k.
18. [HJ85, p. 13] (Rank Inequalities) (Unless specified otherwise, assume that $A, B \in F^{m\times n}$.)
    (a) rank$(A) \leq \min(m, n)$.
    (b) If a new matrix B is created by deleting rows and/or columns of matrix A, then rank$(B) \leq$ rank(A).
    (c) rank$(A + B) \leq$ rank(A) + rank(B).
    (d) If A has a $p \times q$ submatrix of 0s, then rank$(A) \leq (m - p) + (n - q)$.

(e) If $A \in F^{m\times k}$ and $B \in F^{k\times n}$, then

$$\operatorname{rank}(A) + \operatorname{rank}(B) - k \le \operatorname{rank}(AB) \le \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}.$$

19. [HJ85, pp. 13–14] (Rank Equalities)

(a) If $A \in \mathbb{C}^{m\times n}$, then $\operatorname{rank}(A^*) = \operatorname{rank}(A^T) = \operatorname{rank}(\overline{A}) = \operatorname{rank}(A)$.

(b) If $A \in \mathbb{C}^{m\times n}$, then $\operatorname{rank}(A^*A) = \operatorname{rank}(A)$. If $A \in \mathbb{R}^{m\times n}$, then $\operatorname{rank}(A^TA) = \operatorname{rank}(A)$.

(c) Rank is unchanged by left or right multiplication by a nonsingular matrix. That is, if $A \in F^{n\times n}$ and $B \in F^{m\times m}$ are nonsingular, and $M \in F^{m\times n}$, then

$$\operatorname{rank}(AM) = \operatorname{rank}(M) = \operatorname{rank}(MB) = \operatorname{rank}(AMB).$$

(d) If $A, B \in F^{m\times n}$, then $\operatorname{rank}(A) = \operatorname{rank}(B)$ if and only if there exist nonsingular matrices $X \in F^{m\times m}$ and $Y \in F^{n\times n}$ such that $A = XBY$ (i.e., if and only if A is equivalent to B).

(e) If $A \in F^{m\times n}$ has rank k, then $A = XBY$, for some $X \in F^{m\times k}$, $Y \in F^{k\times n}$, and nonsingular $B \in F^{k\times k}$.

(f) If $A_1 \in F^{n_1\times n_1}, \dots, A_k \in F^{n_k\times n_k}$, then $\operatorname{rank}(A_1 \oplus \cdots \oplus A_k) = \operatorname{rank}(A_1) + \cdots + \operatorname{rank}(A_k)$.

20. Let $A, B \in F^{n\times n}$ with A similar to B.

(a) A is equivalent to B.

(b) $\operatorname{rank}(A) = \operatorname{rank}(B)$.

(c) $\operatorname{tr} A = \operatorname{tr} B$.

21. Equivalence of matrices is an equivalence relation on $F^{m\times n}$.
22. Similarity of matrices is an equivalence relation on $F^{n\times n}$.
23. If $A \in F^{m\times n}$ and $\operatorname{rank}(A) = k$, then A is equivalent to $\begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}$, and so any two matrices of the same size and rank are equivalent.
24. (For information on the determination of whether two matrices are similar, see Chapter 6.)
25. [Lay03, Sec. 6.1] If $A \in \mathbb{R}^{n\times n}$, then for any $\mathbf{x} \in \operatorname{RS}(A)$ and any $\mathbf{y} \in \ker(A)$, $\mathbf{x}^T\mathbf{y} = 0$. So the row space and kernel of a real matrix are orthogonal to one another. (See Chapter 5 for more on orthogonality.)

Examples:

1. If $A = \begin{bmatrix} 1 & 7 & -2 \\ 0 & -1 & 1 \\ 2 & 13 & -3 \end{bmatrix} \in \mathbb{R}^{3\times 3}$, then any vector of the form $\begin{bmatrix} a+7b-2c \\ -b+c \\ 2a+13b-3c \end{bmatrix} \left(= \begin{bmatrix} 1 & 7 & -2 \\ 0 & -1 & 1 \\ 2 & 13 & -3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \right)$ is in $\operatorname{range}(A)$, for any $a, b, c \in \mathbb{R}$. Since a row echelon form of A is $\begin{bmatrix} 1 & 7 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$, we know that the set $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 7 \\ -1 \\ 13 \end{bmatrix} \right\}$ is a basis for $\operatorname{range}(A)$, and the set $\left\{ \begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is a basis for $\operatorname{RS}(A)$. Since its reduced row echelon form is $\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$, the set $\left\{ \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is another basis for $\operatorname{RS}(A)$.

2. If $A = \begin{bmatrix} 1 & 7 & -2 \\ 0 & -1 & 1 \\ 2 & 13 & -3 \end{bmatrix} \in \mathbb{R}^{3\times 3}$, then using the reduced row echelon form given in the previous example, solutions to $A\mathbf{x} = \mathbf{0}$ have the form $\mathbf{x} = c \begin{bmatrix} -5 \\ 1 \\ 1 \end{bmatrix}$, for any $c \in \mathbb{R}$. So $\ker(A) = \text{Span}\left(\begin{bmatrix} -5 \\ 1 \\ 1 \end{bmatrix}\right)$.

3. If $A \in \mathbb{R}^{3\times 5}$ has the reduced row echelon form $\begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & -2 & 0 & 7 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$, then any solution to $A\mathbf{x} = \mathbf{0}$ has the form
$$\mathbf{x} = c_1 \begin{bmatrix} -3 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ -7 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$
for some $c_1, c_2 \in \mathbb{R}$. So,
$$\ker(A) = \text{Span}\left(\begin{bmatrix} -3 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -7 \\ 0 \\ 1 \\ 1 \end{bmatrix}\right).$$

4. Example 1 above shows that $\left\{\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 7 \\ -1 \\ 13 \end{bmatrix}\right\}$ is a linearly independent set having the same span as the set $\left\{\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 7 \\ -1 \\ 13 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}\right\}$.

5. $\begin{bmatrix} 1 & 7 \\ 2 & -3 \end{bmatrix}$ is similar to $\begin{bmatrix} 37 & -46 \\ 31 & -39 \end{bmatrix}$ because $\begin{bmatrix} 37 & -46 \\ 31 & -39 \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 7 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}$.

2.5 Nonsingularity Characterizations

From the previous discussion, we can add to the list of nonsingularity characterizations of a square matrix that was started in the previous chapter.

Facts: The following facts can be found in [HJ85, p. 14] or [Lay03, Sections 2.3 and 4.6].

1. If $A \in F^{n\times n}$, then the following are equivalent.
   (a) A is nonsingular.
   (b) The columns of A are linearly independent.
   (c) The dimension of range(A) is n.